14 Saithe in the North Sea, Rockall, West of Scotland, Skagerrak and Kattegat

pok.27.3a46 – *Pollachius virens* in subareas 4, 6, and in Division 3.a

The assessment of saithe in Division 3.a and subareas 4 and 6 follows the protocol and updated reference points defined during the benchmark in February-March 2024 (WKBGAD; ICES, 2024), which updated the model and input methods adopted in the former benchmark and interbenchmark protocol (ICES, 2016; ICES, 2019a).

14.1 General

14.1.1 Stock definition

A summary of available information on stock definition can be found in the Stock Annex.

14.1.2 Ecosystem aspects

No new information on ecosystem aspects was presented at WGNSSK in 2024. A summary of available information, prepared during WKBENCH (ICES, 2011), can be found in the Stock Annex.

14.1.3 Fisheries

A general description of the fishery (along with its historical development) is presented in the Stock Annex.

Saithe are taken mainly in directed trawler fisheries by Norway, Germany, and France. Changes in the fishing pattern of these three fleets began in 2009, but all fleets had largely reverted to their original fishing patterns by 2011 (see Stock Annex for years 2000–2015). For the German and Norwegian fleets, the original fishing pattern is mainly along the shelf edge in Subarea 4 and Division 3.a, while French fleets fish along the northern shelf and west of Scotland (subareas 4 and 6).

The German fleet has been restructured during the last 6 years, and two new vessels entered the fleet while older vessels were taken out. The French fishery is currently at capacity for processing the catch at the vessel; this fishery cannot increase their catches.

The Scottish fleets catch many saithe in mixed demersal fisheries in subareas 4 and 6, that represents an increasing proportion of the total catch (around 17% in Subarea 4 and Division 3.a, and 73% in Subarea 6 in 2023). Discarding continued in 2023 according to data submitted to InterCatch in areas 4, 3.a and 6 despite full landing obligations in place in the EU, UK, and Norway.

In 2021 and 2022, part of the Norwegian TR1 fleet relocated north of 62°N, because of the low North Sea saithe quota and perceived large bycatch of cod (while quota was very low as well) on usual fishing grounds. No such feedback was received from the industry for 2023, and, unlike in the two previous years, the Norwegian quota allocated to trawlers was fished that year.

14.1.4 ICES Advice for the previous year

The information in this section is taken from the 2024 Advice sheet.

Advice for 2025

ICES advises that when the MSY approach is applied, catches in 2025 should be no more than 79 071 tonnes.

The agreed TAC (trilateral agreement) was in line with the ICES advice.

14.2 Management

Changes to the stock assessment and reference points during the benchmark in 2024 (WKBGAD; ICES, 2024), imply a need to re-evaluate the former EU-UK-Norway management strategy to ascertain if it can still be considered precautionary under the new stock perception. Until such a re-evaluation is conducted, or a new management strategy evaluated by ICES is in place, advice will be given according to the ICES MSY approach.

14.3 Data available

14.3.1 Catch

Official landings for each country participating in the fishery, together with the corresponding WG estimates and the agreed international quota ("total allowable catch" or TAC) and ICES estimated discards and BMS landings are presented in Table 14.3.1. England and Wales resubmitted data (catch weights only, no change in sampling data) for the year 2021-2023, and were re-extracted and raised when necessary.

For 2021, only swap of catch weights occurred between métiers aggregated in the same raising groups and had therefore no impact whatsoever. For 2022, the changes were only of a couple kilograms in two strata, unlikely to results in any substantial changes. This was however tested and confirmed using scripts developed during the RDBES-Raise&TAF workshop (ICES, 2023b), but marginal changes in estimates (partially due to difference in rounding used) were carried into the assessment. Only in 2023, were about 220 tonnes added to landings from the TR1 fleet operated by England. InterCatch extractions and raising were therefore performed again, using the same schemes as the previous year, which led to a slight revision of the overall discard rate, from 6.0% to 5.8%, and total catch (SOP age 3+-10) by just +145 tonnes.

In 2024, official landings and ICES estimates were close in both Division 3.a, and subareas 4 and 6. ICES estimates correspond to the sum of products (SOP) uploaded to InterCatch and present a reasonable match for overall catch (101.5%), with an overestimation very close to that obtained in previous year.

In 2024 95% of discards were imported to InterCatch while 5% were raised (Table 14.3.2). Discard observations were not available for some of the fleets landing larger amounts of saithe (Figure 14.3.1). This is mainly the case for the Norwegian fleets. While Norway has a landings obligation policy for all métiers and in all areas, discarding is not monitored and discard information is not collected; therefore, discards for the Norwegian, French, and German trawler fleets (TR1) were raised using provided discard information from the French and German trawler fleets (i.e. targeted saithe fisheries; quarterly stratification). Discards in the trawler fleets (TR1) from other countries were matched for raising by quarter and area (4/6 and 3.a were distinguished); quarter 1 landings in Division 3.a were matched to discard samples in quarter 2, because of missing sampling at the end of the year. Because of lack of sampling data in 2024, all seasons were raised together for all other métiers (all countries) in subareas 4 and 6 together, as was done for data in 2023. For other métiers in Division 3.a, discards were still matched per quarter. Information on discarding from Scottish métiers were not included when raising discards for active gears from other countries because rates were typically high (except, in 2023 and 2024, for TR1 landings declared for the whole year in subarea 6, originating in Scotland and Ireland only, where Scottish discards were used for raising).

The complete time-series of catch, landings, and discards as used in the assessment (age 3-10+) is summarized in Table 14.3.3 and illustrated in Figure 14.3.2. Catch has been relatively stable from 1990 through 2008 and then declined steadily, with a transient increase between 2017–2019, linked to a short increase in quota. The WG estimates of saithe discards (as a proportion of total catch) has remained relatively constant since 2003. Discard estimates were lowest for the period when the saithe trawler fleet changed its exploitation pattern (2009–2011). Prior to 2002, discards were estimated using a constant age-specific discarding rate (see WKNSEA; ICES, 2016). Occasional higher discard rates, particularly in 2016, were due to reported discarding by Scottish fisheries.

Targeted saithe fisheries were covered by the EU Landing Obligation since 2016. Since 2018 saithe is under the landing obligation in all fleets in subareas 4 and 6, and in Division 3.a. Very few BMS landings and no logbook reported discards were reported into InterCatch since 2018 (Table 14.3.2). Sampled and estimated discard rates in 2019–2021 were particularly low, essentially because of very low recruitments in 2018–2021. In 2022, discard rates (all ages) were back to 2018 levels (around 8%), due to larger age 3–4 cohorts, but were down to about 6% in 2023. In 2024, the overall discard rate was estimated at 7.4%, in line with previous years.

14.3.2 Age compositions

International catch data were collated and catch-at-age were generated using InterCatch. Age composition in the landings was based on samples, provided by Denmark, France, Scotland, Germany, Ireland, and Norway, which accounted for 91% of the total landings in 2024 (Table 14.3.4; Figure 14.3.3), back to levels of the years before 2020. This increase in the fraction sampled was mostly due to the French OTB_DEF_>=120_0_0 stratum being sampled again for the first time since 2020. Many fleets do not provide samples for the landings, but these do not usually contribute to a large proportion of the catch. However, the number of samples taken, especially in the targeted trawl fisheries, is an issue (see WKNSEA; ICES, 2016). The quality of the age structure and mean weights-at-age in the catch was however assumed appropriate. Stratification for age compositions was by quarter and area for the unsampled landings, as described in ICES (2016). Since 2021 (data year), however, age composition of landings in area 6 have been matched to all seasons due to lack of sufficient seasonal data. The seasonal partition is prioritized because of the seasonal nature of the fishery, which is mostly driven by seasonality in catches from the Norwegian TR1 fleets. This has shifted over the last decade from a fishery dominated by catches in Q1 and Q2 (targeting mainly spawning fish), to a fishery dominated by catches in Q2 and 3, mostly driven by changes in fishing patterns by the Norwegian TR1 fleet. Smaller and younger fish are generally found in Division 3.a.

Ninety-nine percent of the provided discards (corresponding to 94% of the estimated discards) were sampled for age distributions in 2024 (Table 14.3.4). All age information for discards were from Scotland, and Denmark to a lesser extent (Figure 14.3.4) which also have by far the largest amounts of discards. While the proportion of discards sampled for age distribution was high (Table 14.3.4), the number of age samples per métier is often low (WKNSEA; ICES, 2016). Due to a very uneven spatial and temporal coverage, catch-at-age information was estimated per Subarea for all seasons together. This is however believed not to be a critical issue for the quality of assessment as discards are typically low. Catch-at-age for the BMS landings was generated from the discards age information.

Total catch-at-age data are given in Table 14.3.5, while landing-at-age and discard-at-age data are respectively given in Tables 14.3.6 and 14.3.7. Age 3 fish made up a smaller portion of the landings in recent years (Figure 14.3.5). The last strong year class in the catch appears to be the 2009 year class as seen in the discards in 2012 at age 3 and landings in 2013 at age 4. A slightly stronger year class appears to be entering the discards at age 3 in 2016 and at age 4 in the landings in 2017. Weak cohorts entering the fisheries at age 3 resulted in respective low discards from 2018 to 2021.

14.3.3 Weight-at-age and natural mortality

Weight-at-age from the catch, landing and discard components for ages 3–10+ are presented in tables 14.3.8–14.3.10 and Figure 14.3.6. Catch weights exhibited a downward trend since around 2020 for younger ages, in line with, but less pronounced than, that observed for stock weights (see below). It is particularly marked after 2022, for most ages. The trend over the last 20 years however stays positive for older age classes.

Stock weights for the years 2003–2024 were estimated as model-based weights-at-age using survey data only, because it was shown that catch weights-at-age overestimated stock weights up to about age 6 (WKBGAD; ICES, 2024). The general additive mixed model (GAMM) used includes age, cohort, seasonal and Subarea/Division effects, and implements bias correction for age-based estimates using length stratified sampling as documented in WKMOG (ICES, 2008) Series from all available seasons—(NS-IBTS [G1022 & G2829], SWC-IBTS [G1179 & G4299], IE-IGFS [G7212] and a dedicated Norwegian acoustic and trawl survey (2017–2022)—were used. Weights-at-age are predicted for January per year and Subarea, and weighted means calculated per year over subareas (and age for the plus group), using absolute index-at-age (1–14+) values per Subarea (delta-GAM approach, see Section 14.3.5) as weights. Stock weights-at-age for the year 1967–2002 were estimated as constant proportions of catch weights-at-age, based on ratios estimated for the period 2003–2022. The estimates, as used in the assessment, are presented in Table 14.3.11.

The 2025 update revealed that stock weights-at-age 2024 in the previous assessment were severely overestimated by the intermediate year assumption (last three-year average). It was shown that this could have been avoided by using preliminary estimates based on quarter 1 survey data in the assessment year. Updated methodology for the forecasting of the stock weights-at-age, based on model reviewed and validated at the last benchmark (ICES, 2024), was presented to and validated by WGNSSK 2025. This is described in the working document (Annex 8 Working Document 5): it endorses the use of preliminary estimates based on quarter 1 survey data in the assessment year as intermediate year assumption, and model forecast for the advice year and advice year+1(using the last three-year average proportions of index at age per subarea for the area-weighting).

The update of the series triggered very little revision back in time, compared to the one estimated in 2024 (Figure 14.3.7). There was a decreasing trend in mean weight for ages 5 and older, but that has stopped or been reversed after 2008 (Figure 14.3.7). Weights-at-age for ages 3–7 have been relatively stable, with some variation, over the previous decade (2010’s), but a decrease in weights-at-age has been observed since the beginning of the 2020’s, starting with the younger ages, and propagating to older groups as cohorts grow older. Between 2023 and 2024, a steep decrease in weight-at-age was observed for all age groups. Preliminary data 2025 show a likely stabilisation for age 3-5. The noisy weights-at-age for discards of older age classes during the last decade, reflect a lack of data due to minor discard rates, and should therefore trigger no concern.

Natural mortality-at-age was estimated in the 2024 benchmark based on mean stock weights-at-age, following the equation of Lorenzen (1996), further scaled so that age-9 (≈5 kg) had a natural mortality of about 0.2, as shown in the table below.

Age	3	4	5	6	7	8	9	10+
Natural mortality	0.384	0.335	0.294	0.259	0.232	0.212	0.197	0.177

14.3.4 Maturity

The 2024 benchmark (WKBGAD; ICES, 2024) validated a model-based estimation of proportion mature at age. Because of the documented bias in weight-at-age in catches, only samples from surveys in Q4 and Q1—NS-IBTS (G1022), SWC-IBTS (G1179 & G4299), SCOWCGFS (G4748 & G4815), IE-IGFS (G7212) and a dedicated Norwegian acoustic and trawl survey (2017−2022)—were used. Age and years of the Q4 surveys were forward-shifted to match the closest spawning season. The GAMM—which includes among others quarter and Subarea effects, and implements the same bias correction as mentioned for the stock weights model—was used to make predictions of proportion mature per Subarea, in Q1, for the years 1992−2023. These were further used for population-level inference by calculating weighted means per year over subareas (and age for the plus group), using the absolute index-at-age (1−14+) values per Subarea as weights (see Section 14.3.5). The entire series is updated when a new year of data are added (values used in the assessment are in table 14.3.12): the 2025 update exhibited some downward revision back in time when compared to the estimates in the 2024 assessment (Figure 14.3.8), especially at age 5 and 6, and a very slight revision upward at age 3 in historical data.

The 2025 update revealed some downward revisions of the 2024 estimates for the most abundant age groups compared to the intermediate year assumption (last three-year average). It was shown that this could have been anticipated using estimates based on quarter 1 survey data in the assessment year. Updated methodology for the forecasting of the proportion mature-at-age, based on the model reviewed and validated at the last benchmark (ICES, 2024), was presented to and validated by WGNSSK 2025. This is described in the working document (Annex 8 Working Document 5): it endorses the use of estimates based on survey data up to quarter 1 in the assessment year as intermediate year assumption, and model forecast for the advice year and advice year+1 (using the last three-year average proportions of index at age per subarea for the area-weighting).

14.3.5 Catch per unit effort and research vessel data

Indices used in the final assessment are included in Table 14.3.13. Data for the Norwegian, French, and German commercial trawler fleets were combined into one standardized CPUE index (integrating Year, Month, Country, Power, Vessel effects and spatial and spatio-temporal GRMF, without interactions), which is then tuned to the exploitable biomass (see Stock Annex for details). One fisheries-independent survey index was included for tuning of the assessment; an extended area (including west of Scotland) Q3−4 delta-GAM index (delta-GAM approach: Berg *et al.*, 2014), ages 3–8, 1992–2024 ("Survey index Q3−4"). The same approach was used to update the index per subArea for ages 1−14+, 1992–2024.

The commercial CPUE index exhibited, in 2024, a steep decline when compared to 2023, and was almost back to the all-time low in 2021 (Figures 14.3.9 and 14.3.10). Model diagnostics were very similar to those obtained at the 2024 benchmark, with data up to 2022 (not shown). Only very modest revisions were observed in the previous years of the scaled index (Figure 14.3.10), the model presenting very little retrospective patterns. This new index, based on a spatio-temporal

GAMM (WKBGAD; ICES, 2024), is deemed more appropriate to account for changes in fishing patterns, especially those not driven by saithe abundance and distribution (bycatch avoidance, temporary loss of access to some areas), than the former GLM-based index. Thanks to the area-weighting standardization, it is also expected to be more robust to contraction or extension of the spatial distribution of the population. Spatial patterns in the residuals were reasonable for most years (Figure 14.3.11)—and the other diagnostics were very similar to those obtained during the benchmark (WKBGAD; ICES, 2024; figures not shown). However, for 2024 the spatial predictions reveal that there is a marked difference in CPUEs between the areas in the north/north-west of the stock area and eastern areas along the Norwegian trench (Figure 14.3.9). CPUEs were higher along the Norwegian trench compared to the areas around Fladen ground and towards west of Scotland. Since Norway, France and Germany concentrate their fishing effort in different areas (Figure 14.3.12), the index signal from each fleet must also be different. An exploratory leave-one out analysis using the previously used GLM-based index model confirms that mainly the CPUEs observed for the French fleet fishing in the northern and north-western parts of the stock area lead to the overall low index value for 2024 (Figure 14.3.13). Although the German fleet has increasing CPUEs compared to 2023 within their main fishing grounds along the southern and middle parts of the Norwegian trench, the CPUE index for the full stock area suggests a decline of exploitable biomass between 2023 and 2024.

The survey index decreased in 2024 for ages 4, 7 and 8, while it increased for ages 3, 5 and 6 (Table 14.3.13, Figure 14.3.14). Age three has been overall increasing in the last five years, after reaching an historic low in 2019, but remained at quite low levels. Age-five abundance exhibited a steep increase in 2024, signs of which can be tracked back to a seemingly larger recruiting cohort (age 3) in 2022. Indices at ages over six reached, in 2024, relatively low levels. The index coefficients of variation (CV) were, in 2024, reasonably low for age 3-6, but increased very steeply for age 7 and 8, indicating increased uncertainty for the older fish (Figure 14.3.15). Model diagnostics were very similar to those obtained at the 2024 benchmark, with data up to 2022 (not shown). The model-based index, in particular, exhibited hardly any revision of values in 2023 and before when adding one year of data (Figure 14.3.14).

Figure 14.3.16 shows the updated index per subArea, as absolute sum (over cells in Subarea) of catch predictions used for the weighting of biological parameters among Subareas and ages within age groups. Proportions among Subareas were stable, when compared to previous years (Figure 14.3.17). A more detailed analysis of the spatial distribution of Q3-4 delta-GAM survey index predictions confirms a concentration of the saithe stock in 2024 along the southern and middle parts of the Norwegian trench, while especially areas towards the north-west show much lower densities (Figure 14.3.18). Therefore, spatial patterns observed in the scientific surveys and for the commercial CPUE index are consistent (see above).

14.4 Data analyses

14.4.1 Exploratory survey-based analyses

Numbers-at-age for saithe ages 3 to 8 (Survey index Q3–4) on the log–scale, linked by cohort, showed year effects (for example, consistently high values around 2016; Figure 14.4.1, top-left panel), but not as marked as they used to be with the former design-based index (WGNSSK; ICES, 2023a). The ability to track cohorts has been slightly diminished in last 20 years of the survey (Figure 14.4.1, top right panel), but again, has improved compared to the former design-based index (WGNSSK; ICES, 2023a). The survey catch-numbers correlate poorly between cohorts for ages 3 and 4, but are stronger for subsequent ages (Figures 14.4.1, top-right panel, and 14.4.2). This is likely because age 3 fish are not consistently fully represented in the survey (“hook” patterns at age 3 in the abundances of some cohort: Figure 14.4.1, bottom-left); fish begin migrating out of the inshore nursery areas at age 3, but do not fully recruit to the more oceanic population (and fishery) until after age 5. The new delta-GAM approach however appears to have improved representativeness for age 4 (as revealed by better internal consistency with older ages), but also age 3, which has now more consistent hook patterns in recent years.

Despite the improvements brought by the delta-GAM approach (WKBGAD; ICES, 2024), this survey index remains inherently noisy. A high degree of uncertainty in the survey index has been commented on previously (WKNSEA; ICES 2016), and some observations remain relevant to the new index, among which the lack of sampling of un-trawlable areas on the northern part of the shelf where dense aggregations are common, or—to a lesser extent than with the former design-based index—the strong influence of occasional very large single samples. Despite this, the index is still considered the best available information for young saithe, although the assessment places more weight on the CPUE index (estimated observation variance 0.08 vs. 0.48 for the survey index; also highlighted by the leave-one-out analyses in former years; see Section 14.4.4). Survey index Q3–4 values used in the final assessment are in Table 14.3.13.

14.4.2 Exploratory catch-at-age-based analyses

The outcome of WKNSEA 2016 was to remove the 3 CPUE series for the targeted trawl fisheries, partially due to concerns over using information in the catch-at-age matrix in both the CPUE and in the catch-at-age and because more weight was given to 3 indices within the former assessment model (artificially giving higher weighting to the CPUE indices). A standardized combined CPUE index was created for the French, German, and Norwegian trawl fleet targeting saithe, which was then tuned to the exploitable biomass, removing the need to use the information in the catch-at-age matrix twice (see WKBGAD for details; ICES, 2024).

The fit of the CPUE to the exploitable biomass shows limited ability to render annual variations but reflects well the index main trends (Figure 14.4.3). The recent update to an area-weighted index, based on a model with spatio-temporal GMRF, with a vessel random effect, is expected to make it more robust to changes in the spatial distribution of the effort and/or resource, as well as a possible drift in fishing strategy and experience. It however cannot be discounted that, in particularly if fishers are very good at avoiding areas with low density of saithe, the index might lack capacity to properly render the repartition of those and generate an overestimation in periods of low abundance.

14.4.3 Assessments

The assessment of North Sea saithe was carried out using a state-space stock assessment model (SAM; Nielsen and Berg 2014; Berg and Nielsen 2016). This is the first update following revision of the assessment model during a benchmark (WKBGAD; ICES, 2024). Settings used in the final assessment are given in Table 14.4.1.

14.4.4 Final assessment

Estimated fishing mortality-at-age are given in Table 14.4.2 and Figure 14.4.4. F for age 3 has declined drastically from 1990 and is now close to 0.1, while F for the older age classes has also decreased slightly over the same period. The change in F at age 3 occurred when the catches in the purse-seine fishery declined. Age 4 also exhibited a similar decrease in fishing mortality, around the same time, and stabilized around 0.25. For ages 5+, catchability shows a dome shaped pattern, with highest catchability for age 6 in recent years (Figure 14.4.4, right panel). With the lower fishing mortalities up to 2016, fish have been allowed to increase in size (and age) and are likely targeted more than the younger age classes up to age 4 (as observed in Figure 14.4.4).

Fishing mortality, after 2016, has however increased again for age classes 4+, followed by a decrease in 2020–2022. Recruitment was also very low from 2018 to 2021. Estimated population numbers-at-age are in Table 14.4.3.

The survey index-at-age fits are shown in Figure 14.4.5. While accounting for the correlation between ages within years, the IBTS–Q3 residuals show little patterns (one-step ahead residuals, Figure 14.4.6). A slight trend appears in the catch residuals at age 6–9 in recent years of the series, revealing slight systematic overestimations. This might be due to conflicting signals borne by the CPUE index (tuned to the exploitable biomass) and the survey index, or decreasing catchability of older fish by the survey. The strength of the correlation between subsequent ages in survey residuals, is strong for all ages in the survey index ($\varrho = 0.83$ for successive ages 3–7, and $\varrho = 0.74$ between age 7–8; Figure 14.4.7, right panel). For catch observations, the correlation between successive ages was low between ages 3–6 ($\varrho = 0.15$), while it was more pronounced between ages 6–10+ ($\varrho = 0.72$; Figure 14.4.7, left panel), which can be the sign of correlated errors in catch-at-age estimates. Process residuals (Figure 14.4.8) do not exhibit obvious trends.

The retrospective analysis shows a very slight overestimation of SSB for all peels, while F had no obvious pattern. Most of the retrospective in recruitment (Mohn's rho = −0.21) was due to a revision upward of the recruitment in 2022 (Figure 14.4.9). Although SSB tends to be slightly overestimated (Mohn's rho = 0.06), all peels for SSB fell within the confidence intervals of the most recent assessment. For F, the Mohn's rho was low (0.02). The systematic underestimation of F by the model used in the previous years was not observed anymore after the model update (WKBGAD; ICES, 2024), and all peels fell within the now narrower confidence intervals of the final year's assessment.

The leave-one out and final assessment results are in Figures 14.4.10 and 14.5.1 respectively. The two series yielded different trends in the present assessment for SSB and F. Without the CPUE index, the model estimated a lower final SSB, which resulted in higher estimated fishing mortality. This contrast to the former model and inputs, which tended to produce less conservative assessment without the CPUE index. Recruitment estimates, as usual, are little affected by the choice of data.

14.5 Historic stock trends

The historic stock and fishery trends from the final assessment are presented in Figure 14.5.1 and Table 14.5.1. Because of the benchmark in 2024, the historic perception of the stock has changed when compared assessment years before 2024. This is mostly due to revision of the reference points, with higher B_{lim} and B_{pa} and lower F_{MSY} and F_{MSY} range, while trends and final year estimates of SSB and F have hardly changed (WKBGAD; ICES, 2024). Recruitment has been low and highly variable since 1990. Both 2015 and 2016 show slightly higher recruitment than the average of the last ten years, while 2018–2023 (except for 2022, again higher than the last 10-year average) were the lowest estimates for the whole time-series. SSB, has fluctuated around 195 000 tonnes in the 2010s, which is below the average of the 2000s (around 234 000 tonnes). Short-term variations exhibit a decline in SSB since 2018, and was down to about 10 000 tonnes below B_{lim} in 2024. Survivors from 2024 suggest a SSB in 2025 back to about 5000 tonnes above B_{lim} (also dependent on short-term forecast assumptions for the biological estimates and recruitment 2025, which amount to about 8% of the estimated SSB in 2025). Fishing mortality has generally declined since the mid–1980s but has exhibited a distinct raise from 2016 to 2019. It has since decreased again and is now estimated to lie just above F_{MSY} since 2022.

14.6 Recruitment estimates

Currently, no independent survey provides an estimate of incoming recruitment. The resampling among 2015–2024 Age-3 estimates (with a geometric mean about 94 million individuals) used in the short-term forecast is a conservative assumption taking into account recent low recruitment.

14.7 Short-term forecasts

A short-term forecast was carried out based on the final assessment.

Weight-at-age in the stock were preliminary model estimates for the intermediate year, and model-based projections for the following years (ref to WD). Catch and landing weights-at-age in the forecast were the forecast stock weights times the last 3-year average of the catch(resp. landing)/stock weights ratios (at age). The exploitation pattern (selectivity pattern) was chosen as the mean exploitation pattern over the last three years scaled to $F_{4–7}$ in 2024. The fishing mortality in the intermediate year was based on a F *status quo* assumption ($F_{4–7,\ 2025} = F_{4–7,\ 2024} = 0.338$) as catches in 2024 were well below the agreed TAC (60 kt vs. 79 kt) and the estimated F_{2024} fell reasonably close to F *status quo* assumption used in 2024 (0.338 vs. 0.320) (table 14.7.1), and the situation in 2025 was expected to be similar. A TAC constraint 79 071 tonnes, $F_{4–7,\ 2025} = 0.45$; Council Regulation, 2025), would have led to unrealistically high catches considering the quota undershoot the year before. Population numbers-at-age for ages 4 and older in 2025 were survivor estimates, while numbers-at-age 3 were resampled from the past 10 years (2015–2024). The short-term projection was run in SAM.

The intermediate year assumptions for the short-term forecast are given in Table 14.7.2. The options above result in catches in 2025 of 62 705 tonnes and a SSB in 2026 (2025 survivors plus STF recruitment assumption) of 162 097 tonnes, still below MSY $B_{trigger}$ (180 770 tonnes), in the rise compared to 2025 (195 899 tonnes forecasted in 2024; 135 337 tonnes estimated in the current assessment). The large discrepancies between previous STF assumptions and current estimates of SSB led to a revision of the STF assumption for biological characteristics in 2025 (Annex 8 Working Document 5). Reference points and their technical basis are in Table 14.7.3.

The management options are given in Table 14.7.4. Because reference points were re-estimated during the 2024 benchmark and because of Brexit, the management plan for this shared stock (EU, Norway and the UK—as of early 2021) is no longer in use (a new EU-Norway-UK management plan is under discussion); therefore, the MSY approach is used as the basis for advice. The total catch in 2026 is advised to be no more than 60 167 tonnes, where corresponding landings are 56 835 tonnes; this is a 24% decrease when compared to the advised total catch in 2025. More catch options can be found in Table 14.7.4.

Reasonable revisions, before 2024, of the stock weights (upward for older ages in 2023; Figure 14.7.1), proportions mature (downwards, age 3-5; Figure 14.7.2) and number estimated at age (mostly downward, except for age 3; Figure 14.7.3), together resulted in a limited decrease in estimated SSB at age 4-10+ estimates, across all recent years, and a sizeable increase in estimated SSB at age 3 (Figure 14.7.4). Meanwhile, because of the decrease in estimated numbers-at-age while catch estimates were mostly unchanged, the fishing mortality was revised upwards (Figure 14.7.5). The large decrease in the advice, however essentially originated in very distinct overestimations in the 2024 intermediate year assumption, of both stock weights (Figure 14.7.1) and proportions mature (Figure 14.7.2), for those year-classes contributing the most to the SSB (Figure 14.7.6). This overestimation reached up to 60% for age 6 (Figure 14.7.4). The F assumption was still reasonable (slightly underestimated in 2024 – $F_{bar\ 4\text{-}7} = 0.34$ vs. 0.32; Figure 14.7.5), despite lower catch weights-at-age than assumed (not shown), thanks to the substantial TAC undershoot. Similar misestimations were also revealed for the 2025 TAC year estimates in 2024, when compared to the 2025 intermediate year assumption in 2025 (now using more realistic biological estimates, based on preliminary data up to Q1 of the assessment year).

Retrospectively, the exploitation pattern assumptions were quite sensible, as hardly any changes in scaled F were observed (Figure 14.7.7).

The contributions of the 2017–2023 year classes to landings in 2026 are shown in Table 14.7.5. The 2019–2022 year classes are expected to contribute the most to the landings in both numbers and weight. The relatively large 2019 cohort is expected to continue having largest contribution to landings, despite reaching age 7 (unlike previous assessments, where age seven was less prominent). Recruitment-at-age 3 in 2026 is not expected to contribute greatly to the catches in 2026; rather, ages 4–7 are the main contributors (about 75% of projected landings for 2026).

14.8 Medium-term and long-term forecasts

No medium-term or long-term forecasts were carried out.

14.9 Quality and benchmark planning

The assessment was revised during a benchmark in 2024. Some former issues were not addressed and remain relevant, while some considerations for future research arose and are now listed below.

14.9.1 Quality of the assessment and forecast

Conflicting signals between the survey and fishable biomass index contributes to the assessment uncertainty and a (reasonable) retrospective pattern observed.

The fraction of fish at age 3 migrating into the survey area (and the fishery) is low and varying between years with no obvious trend. Observations of saithe at age 3 are not suitable for predicting year-class strength. This means that estimated recruitment values in the final assessment year are highly uncertain. Estimates of recruitment for a given year class tend to be revised considerably with successive assessments (see Section 14.4.4).

Retrospectively, the short-term forecast assumptions used until 2024 were not robust to strong trends in biological estimates, and resulted in large overestimation of the SSB 2024 (survivors) in the 2024 assessment. WGNSSK 2025 therefore adopted a new methodology, based on preliminary model fit (using quarter 1 data in the assessment year) and model projections to provide more realistic biological estimates of stock, catch and landing weights, and proportion mature, for the STF (Annex 8 Working Document 5).

14.9.2 Issues for future benchmark

14.9.2.1 Data

Stock definition

The North Sea saithe stock is influenced by migrations to and from the North Sea. This can potentially lead to the observed year effects in survey indices. It needs to be analysed if the inclusion of spawning grounds north of 62°N could improve the assessment. A tagging study (IMR) is underway and may help inform on this issue. The question of whether the population in ICES subareas 7−10 is connected to the North Sea stock, and should be merged to it, was also raised by the last benchmark (WKBGAD; ICES, 2024), that concluded that there is weak, though somewhat inconclusive, evidence that the population west of Ireland is derived from the North Sea stock. It is however unclear if west of Ireland is a sink population or if it contributes to the spawning of the overall stock, and the benchmark recommended that the matter should be evaluated by SIMWG.

Biological sampling

The 2024 benchmark pointed out possible effects on biological parameters estimates of the lack of biological sampling along Norwegian coastal areas (in and east of the Norwegian trench)—where younger and possibly less mature fish reside—that should be examined. This is believed to be potentially more critical for maturity estimates but might also affect stock weights-at-age estimates.

Commercial CPUE index

A significant relationship between commercial CPUE residuals and the percentage of saithe in the catch, persisted after revision of the index model during the 2024 benchmark. This extra information at the catch operation level (likely variable spatially, and not definable in each cell of the prediction grid) cannot be used as a covariate in the context of an area-weighted index. It could however potentially be co-modelled with the CPUE, and help better predict very large and/or very small catch rates, which remain poorly fitted, and result in heavy-tailed residual distribution.

Recreational catches

Although not expected to be important within the North Sea, the magnitude of recreational saithe catches should be formally investigated to inform the assessment quality. In the context of uncertain geographical stock limits, the NS saithe dynamics might, for instance, be influenced by comparatively more important recreational fisheries in the Norwegian Sea.

14.9.2.2 Assessment

The assessment model was revised in the 2024 WKBGAD benchmark (ICES, 2024). The group pointed out that, despite the model behaving appropriately, the correlation structure in catch observation across ages over 6-year-old was somewhat unusual and hinted at correlated errors in catch-at-age. It therefore recommended disentangling the sources of such assumed correlated errors, or the identification of any other process that could generate the observed correlation in catch residuals across ages.

14.9.2.3 Forecast and reference points

Forecast

The SAM forecast assumption for recruitment is based on resampling from historical recruitment values from a defined number of historical years. Depending on the time-series, this may result in a bimodal distribution for the assumed recruitment in forecasted years. Forecasted numbers (and SSB) are likely to be smoother in their distribution due to forecast stochasticity, but the effect of this behaviour on advice should be investigated further. Use of a geometric mean of historical recruitment is not currently possible in SAM but could be suggested in order to reduce this effect. Adequacy and implementability of recruitment forecasted as a random effect with autocorrelation (RW, AR1…), was not considered at the last benchmark, but should also be evaluated in future.

The setting of a random seed value is important for comparing between forecast scenarios. Forecast scenarios involving a prescribed F had consistent median recruitment; however, scenarios that solve for an F that results in a given stock size (e.g. $SSB_{2026} = B_{pa}$ or B_{lim} scenarios), which involve a further iteration process with additional random number generation, resulted in different median recruitment values. This is a reporting issue that arise from instability of the median value resampled from an even number of values (while a reported geometric mean would be more stable, and often more informative). It does not affect the quality of the assessment, only the consistency of reported figures. We have therefore made the choice, since the 2020 assessment, to report the geometric mean of resampled recruitments values in the forecast assumption (not to be mistaken for the use of a geometric mean as point estimate in the forecast).

14.10 Status of the stock

Fishing pressure on the stock is above F_{MSY} and spawning-stock size is below MSY $B_{trigger}$, and between B_{pa} and B_{lim}. That is considering SSB in 2025, which now includes the STF assumption on recruitment in 2025 (contributing to about 8% of SSB), as the proportion mature for age 3 is around 21% (model estimate using Q1 2025 survey data), following update of the maturity ogive during the 2024 benchmark.

14.11 Management considerations

The assessment is sensitive to relatively small changes in the input data. Because this stock suffers from 'poor data', the assessment is relatively uncertain. Recruitment is currently at a low level, and it appears that strong recruitment pulses are more sporadic than in the past.

The reported landings have been relatively stable since the early 1990s; with the exception of the last three years, when a drastic reduction occurred. Landings have been substantially lower than the TAC between 2017 and 2021, despite consecutive reductions in the quota. Since 2022, the TAC is taken again.

Bycatch of other demersal fish species does occur in the target trawl fishery for saithe. Saithe is also taken as unintentional bycatch in other fisheries, and discards do occur.

14.11.1 Evaluation of the management plan

Because reference points were re-estimated during the 2024 benchmark (and before that following an interbenchmark in 2019), the management plan is no longer valid. New EU/Norway management strategies have been proposed and evaluated (ICES, 2019b). However, these evaluations also are now outdated and would need to be at least re-evaluated.

14.12 References

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14.13 Tables and Figures

Table 14.3.1. Saithe in subareas 4 and 6 and in Division 3.a. Official nominal landings (tonnes) of saithe by nation, along side TAC and ICES estimates for the period 2009–2024. ICES estimates are landings reported to ICES and the Working Group.

Subarea 4 and Division 3.a																
Country	**2009**	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
Belgium	27	15	2	2	3	5	6	16	15	14	6	5	11	5	7	4
Denmark	8802	8018	6331	5171	5695	4913	4512	4084	5690	7017	5275	3777	2946	2280	2261	2514
Faroe Isl.	841	146	2	8	3	1	0	18	16	4	14	28	0	<1	1	0
France	7203	4582*	13856*	14093*	8475	7910	11574	10794	10334	12598	11361	9487	9133	9778	11534	9286
Germany	13410	11193	10234	8052	9690	8602	7954	6279	7943	7952	7055	6858	4460	4280	4152	4423
Greenland	927	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Ireland	1	0	0	0	0	0	0	0	0	0	<1	4*	0*	1*	0	0
Lithuania	0	0	0	0	0	0	0	0	0	0	0	3	0	0	5	0
Netherlands	16	3	24	34	168	43	75	112	191	267	178	181	140	51	127	138
Norway	57708	52712	46809	33288	35701	37519	35631	31596	49580	38787	50306	39558	20442	20841	28066	27448
Poland	988	654	584	0	0	0	0	0	0	0	0	0	0	0	0	0
Portugal										0	0	0	0	0	0	0
Russia	13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Subarea 4 and Division 3.a																
Country	**2009**	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
Sweden	1363	1545	1335	1306	1402	1329	1156	1198	1186	1316	1413	1181	1001	1087	1082	1184
UK (E/W/NI)	12584**	11887**	10250**	7287**	10379**	687	8888**	8561**	8640**	12575**	11868**	8657**	8037**	6748**	7788**	8143**
UK (Scotland)						7686										
Total reported	103883	90755	89427	69241	71516	68695	69796	62658	83594	80531	87476	69739	46150	45071	55023	53139
Unallocated	2090	6012	2101	1623	−110	677	−393	-154	−2024	1335	173	119	797	1076	1857	960
BMS landings									< 1	11	20	10	4	4	6	8
ICES estimate[#]	105973	96767	91528	70864	71406	69372	69403	62504	81570	81866	87649	69858	46947	46147	56880	54099
TAC	125934	107000	93600	79320	91220	77536	66006	65696	100287[##]	105793[##]	93614	79813	59512	44950	53374	66876

*** Official values are preliminary.**

**** Scotland+E/W/NI combined.**

 # Since 2016, landings include the Norwegian component of BMS landings.

Includes top-up (4.1% in 2017, 12.57% in 2018).

Table 14.3.1. Saithe in subareas 4 and 6 and in Division 3.a. Official nominal landings (tonnes) of saithe by nation, along side TAC and ICES estimates for the period 2009–2024. ICES estimates are landings reported to ICES and the Working Group. (continued)

Subarea 6																
Country	**2009**	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
Denmark	0	0	0	0	0	20	0	0	5	1	7	0	0	0	< 1	< 1
Faroe Islands	60	24	5	6	25	29	3	7	13	21	7	3	2	0	< 1	< 1
France	2102	2008	2357	2612	3814	2904	3484	2299	3968	3626	1336	1263	1316	1217	547	511
Germany	298	257	0	9	0	0	0	9	< 1	< 1	0	0	1	0	< 1	0
Ireland	407	520	359	364	313	128	105	185	171	231*	109*	125*	72*	133*	115	39
Netherlands	0	0	0	0	0	0	6	12	3	100	4	< 1	3	0	10	0
Norway	68	121	240	5	715	442	677	555	633	955	478	1	0	< 1	1	1
Poland													1	0	0	0
Russia	4	2	0	0	0	9	1	0	2	0	2	0	0	0	0	0
Spain	8	18	31	13	21	9	15	15	4	7	24	15	38	21	5	1
Sweden	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
UK (E/W/NI)	3491**	3168**	4500**	4549**	3646**	97	3286**	2770**	2652**	2764**	2824**	2666**	2001**	2050**	1583**	1575**
UK (Scotland)						3191										
Total reported	6438	6118	7492	7558	8534	6829	7577	5852	7453	7706	4791	4074	3434	3421	2262	2128

Subarea 6																
Country	**2009**	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
Unallocated	−144	145	−575	−9	119	191	−43	−279	−337	−1065	84	7	100	112	78	-33
BMS landings									0	31	< 1	< 1	< 1	< 1	<1	0
ICES estimate#	6294	6263	6917	7549	8653	7020	7534	5573	7116	6641	4875	4081	3534	3533	2340	2095
TAC	13066	11000	9570	8230	9464	8045	6848	6816	10404##	10215##	9713	8280	6175	4664	5538	6939

*** Official values are preliminary.**

**** Scotland+E/W/NI combined.**

\# **Does not include BMS landings (no Norwegian BMS landings from subarea 6).**

\## **Includes top-up (4.1% in 2017, 4.76% in 2018).**

Table 14.3.1. Saithe in subareas 4 and 6 and in Division 3.a. Official nominal landings (tonnes) of saithe by nation, along side TAC and ICES estimates for the period 2009–2024. ICES estimates are landings reported to ICES and the Working Group. (continued)

Subareas 4 and 6 and Division 3.a																
	2009	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
ICES estimate#	112267	103030	98446	78414	80059	76392	76936	68077	88686	88507	92524	73938	50482	49680	59221	56194
TAC	139000	118000	103170	87550	100684	85581	72854	72512	110691##	116008##	103327	88093##	65687	49614	58912	73815

\# **Since 2016, landings include Norwegian component of BMS landings.**

\## **Agreed upon TAC, including landings top-up.**

Table 14.3.2. Saithe in subareas 4 and 6 and in Division 3.a. Catch data (2024; all ages, not the sum over products for ages 3–10+ used in the assessment) imported into InterCatch and percentage of sampling strata for discards raised within InterCatch.

Catch Category	Raised or Imported	2024	
		Weight (tonnes)	Percentage
BMS landing	Imported data	2.6	100
Discards	Imported data	4203	95
Discards	Raised discards	214	5
Landings	Imported data	55317	100
Logbook registered discard	Imported data	0	0

Table 14.3.3. Saithe in subareas 4 and 6 and in Division 3.a. Working Group estimates of catch components by weight (t) for ages 3–10+, as used in the assessment. Norway was under landings obligations since 1988, but records are unclear whether saithe was fully in the landing obligation from that time.

Year	Catches	Landings[#]	BMS Landings	Discards[##]	Percentage discards
1967	101331	88339		12992	13
1968	134559	113741		20818	15
1969	150293	130580		19713	13
1970	270829	235012		35817	13
1971	309177	265356		43821	14
1972	296481	261914		34567	12
1973	275164	242513		32651	12
1974	337021	298347		38674	11
1975	304645	271610		33035	11
1976	423347	343898		79449	19
1977	239913	216393		23520	10
1978	176851	155124		21727	12
1979	142647	128352		14295	10
1980	145289	131897		13392	9
1981	148244	132273		15971	11
1982	202111	174336		27775	14
1983	203018	180040		22978	11
1984	240566	200843		39723	17

Year	Catches	Landings[#]	BMS Landings	Discards[##]	Percentage discards
1985	273672	220870		52802	19
1986	232795	198605		34190	15
1987	192380	167503		24877	13
1988	154252	135176		19076	12
1989	124599	108892		15707	13
1990	124450	103831		20619	17
1991	130973	108071		22902	17
1992	115537	99745		15792	14
1993	132618	111499		21119	16
1994	126759	109621		17138	14
1995	141190	121795		19395	14
1996	128896	114968		13928	11
1997	120103	107348		12755	11
1998	117222	106126		11096	9
1999	119467	110531		8936	7
2000	93795	85781		8014	9
2001	102859	91741		11118	11
2002	129847	110911		18936	15
2003	121656	110282		11374	9
2004	113792	107356		6436	6
2005	121217	118625		2592	2
2006	128711	120414		8297	6
2007	106333	94958		11375	11
2008	129887	121618		8269	6
2009	114520	110972		3548	3
2010	104723	102128		2595	2
2011	102006	98034		3972	4
2012	87049	78144		8905	10
2013	87271	79859		7412	8

Year	Catches	Landings#	BMS Landings	Discards##	Percentage discards
2014	82172	76057		6115	7
2015	81445	76748		4697	6
2016	77672	67620	0	10052	13
2017	94581.5	88010	0.5	6571	7
2018	95447	88328	42	7076	7
2019^	96634	92390	20	4224	4
2020	76820	73791	10	3019	4
2021	50951	50124	0.4	827	1.6
2022	53808	49558	0.2	4250	8
2023	61041	57795	0.6	3246	5
2024	60303	56074	2.6	4229	7

 # Since 2016, landings include the Norwegian component of BMS landings.

Since 2016, discards minus BMS landings from EU fleets officially reported in logbooks.

^ Includes 937 tonnes of missing Swedish landings and corresponding 109 tonnes of discards (based on discard rate estimated in division 4.a).

Table 14.3.4. Saithe in subareas 4 and 6 and in Division 3.a. Amount (weight and percentage) of sampled or estimated age distributions of catch data (2024) imported or raised in InterCatch. Weight in tonnes corresponds to the catch in tonnes imported for all ages, and not to the SOP used in the assessment for ages 3–10+.

Catch Category	Raised or Imported	Sampled or Estimated	2024	
			Weight	Percentage
Logbook Registered Discard	Imported Data	Estimated Distribution	0	0
Landings	Imported Data	Sampled Distribution	50489	91
Landings	Imported Data	Estimated Distribution	4828	9
Discards	Imported Data	Sampled Distribution	4165	94
Discards	Raised Discards	Estimated Distribution	214	5
Discards	Imported Data	Estimated Distribution	38	1
BMS landing	Imported Data	Sampled Distribution	0	0
BMS landing	Imported Data	Estimated Distribution	2.6	100

Table 14.3.5. Saithe in subareas 4 and 6 and in Division 3.a. Catch numbers (thousands) at age for the age range used in the assessment.

Year/Age	3	4	5	6	7	8	9	10+
1967	26948	19395	16672	2358	1610	299	203	185
1968	36111	25387	14153	6166	433	247	127	147
1969	47014	21142	11869	7790	5795	810	642	151
1970	57920	91668	16102	12416	3932	1834	326	270
1971	108549	69105	35143	4848	4290	2910	1922	782
1972	74755	79033	27178	21711	3709	3014	1682	1625
1973	84484	45078	28822	16443	8511	2047	1391	2407
1974	104086	40345	15160	21179	14810	5321	1514	1977
1975	88613	30927	11077	7746	13792	9577	3591	2717
1976	323156	63447	12556	6401	4016	5488	3678	3528
1977	42701	65727	15839	5620	3814	3528	3909	4753
1978	54515	32608	19389	3390	1149	1057	788	3522
1979	25395	16999	12004	8906	2833	750	554	2112
1980	27203	14757	9677	6878	5714	1177	522	2327
1981	40705	9971	7235	3763	3368	3475	674	2564
1982	49595	48533	9848	6120	2166	1489	1007	1268
1983	43916	24637	27924	5813	4942	1529	1062	1342
1984	125848	38470	13910	13320	1673	1281	344	653
1985	208401	66489	14257	4878	3034	698	409	750
1986	86198	109080	16302	5509	2629	1490	457	910
1987	48545	116551	15019	3233	1829	1269	933	707
1988	50657	31577	37919	3918	1927	1130	796	687
1989	34408	36772	14156	11211	1572	757	430	493
1990	63454	23416	12154	4826	2803	762	288	368
1991	71710	35719	8016	3669	1733	976	376	463
1992	28617	40193	13691	3269	1539	712	531	426
1993	58813	24905	12715	3199	1583	1547	835	1037
1994	31034	48062	13992	4399	957	354	438	803

Year/Age	3	4	5	6	7	8	9	10+
1995	41461	31130	15884	3864	3529	690	566	809
1996	17208	46468	12653	7915	3194	827	215	496
1997	23380	23077	32395	3763	2666	1036	299	292
1998	16113	37088	17570	16459	2253	1234	581	280
1999	14661	16588	28645	8588	10169	2401	914	665
2000	10985	20680	9597	12632	3190	3302	657	446
2001	24961	21100	24068	3429	3621	1814	1655	248
2002	17570	37489	14736	13731	2309	2544	1321	1575
2003	28296	31752	20631	6836	6855	1535	2000	2042
2004	13642	24479	15649	15220	2037	2164	1300	1066
2005	12690	15473	19060	20042	7956	1628	1188	1151
2006	17313	31972	10381	11286	8395	3824	1008	1281
2007	24614	13314	20919	7175	5564	3610	1218	930
2008	7620	30911	12540	14941	5088	3285	3551	3118
2009	7438	15507	14222	5847	8512	2994	1519	2945
2010	8766	9249	9440	6511	2671	4773	1679	2707
2011	12786	24269	8980	3674	2867	1208	1564	3877
2012	14334	13053	16948	4075	1977	1268	541	2611
2013	7267	30318	5312	7869	1890	1241	616	1658
2014	4055	14322	15195	3957	4124	1040	429	1389
2015	8369	8323	14259	8254	1862	1623	715	977
2016	7382	14241	9661	5729	2758	1430	853	1317
2017	4977	18989	9773	6247	5364	1876	820	1113
2018	2603	16250	18858	7376	2142	2027	978	1178
2019	6240	8570	14841	10394	2881	1127	1027	1236
2020	2511	11823	7627	7436	4246	967	381	627
2021	6129	6236	4910	2899	2924	1061	347	695
2022	4788	7701	4806	3499	2046	1262	725	582
2023	6374	18694	7803	2570	1863	826	676	815

Year/Age	3	4	5	6	7	8	9	10+
2024	7496	10270	16392	4062	1128	749	396	584

Table 14.3.6. Saithe in subareas 4 and 6 and in Division 3.a. Landings numbers (thousands) at age for the age range used in the assessment.

Year/Age	3	4	5	6	7	8	9	10+
1967	17330	16220	15531	2303	1594	292	198	183
1968	23223	21231	13184	6023	429	242	123	145
1969	30235	17681	11057	7609	5738	791	626	150
1970	37249	76661	15000	12128	3894	1792	318	267
1971	69808	57792	32737	4736	4248	2843	1874	774
1972	48075	66095	25317	21207	3672	2944	1641	1607
1973	54332	37698	26849	16061	8428	2000	1357	2381
1974	66938	33740	14123	20688	14666	5199	1477	1955
1975	56987	25864	10319	7566	13657	9357	3501	2687
1976	207823	53060	11696	6253	3976	5362	3586	3490
1977	27461	54967	14755	5490	3777	3447	3812	4701
1978	35059	27269	18062	3312	1138	1033	768	3484
1979	16332	14216	11182	8699	2805	733	540	2089
1980	17494	12341	9015	6718	5658	1150	509	2302
1981	26178	8339	6739	3675	3335	3396	657	2536
1982	31895	40587	9174	5978	2145	1454	982	1254
1983	28242	20604	26013	5678	4893	1494	1036	1327
1984	80933	32172	12957	13011	1657	1252	335	646
1985	134024	55605	13281	4765	3005	682	399	742
1986	55435	91223	15186	5381	2603	1456	445	900
1987	31220	97470	13990	3158	1811	1240	910	700
1988	32578	26408	35323	3828	1908	1104	776	680
1989	22128	30752	13187	10951	1557	739	419	488
1990	40808	19583	11322	4714	2776	745	281	364
1991	46117	29871	7467	3583	1716	953	367	458

Year/Age	3	4	5	6	7	8	9	10+
1992	18404	33614	12753	3193	1524	696	518	422
1993	37823	20828	11845	3125	1568	1511	814	1026
1994	19958	40193	13034	4297	947	346	427	794
1995	26664	26034	14797	3774	3494	674	552	800
1996	11066	38861	11786	7731	3163	808	210	491
1997	15036	19299	30177	3676	2640	1012	291	288
1998	10363	31017	16367	16077	2231	1206	567	277
1999	9429	13872	26684	8389	10070	2346	891	657
2000	7064	17295	8940	12339	3159	3226	641	441
2001	16052	17646	22421	3349	3586	1772	1614	245
2002	9131	31779	12286	13307	2245	2220	1199	1479
2003	13009	24646	20397	6836	6855	1535	2000	2042
2004	8037	20071	15649	15220	2037	2164	1300	1066
2005	9191	15473	19060	20042	7956	1628	1188	1151
2006	12200	26690	9986	11286	8395	3824	1008	1281
2007	15181	10163	19157	7078	5564	3610	1218	930
2008	6924	23230	10930	14196	4977	3276	3551	3118
2009	6607	14349	13827	5817	8419	2978	1505	2934
2010	7880	8859	9174	6394	2670	4762	1679	2669
2011	10150	22799	8852	3630	2860	1183	1563	3869
2012	7029	11712	15572	4016	1971	1267	537	2610
2013	4999	25516	4974	7645	1886	1241	616	1658
2014	3099	12117	13380	3737	4047	1036	429	1388
2015	6206	7392	13555	8021	1844	1621	715	975
2016	3508	10374	8756	5156	2732	1423	852	1317
2017	3033	15139	8795	6179	5362	1876	820	1111
2018	2017	12994	16936	7043	2125	2016	976	1177
2019	5456	8125	13826	9797	2842	1116	1025	1235
2020	1997	10870	7243	7326	4113	959	377	619

Year/Age	3	4	5	6	7	8	9	10+
2021	5693	6073	4859	2863	2924	1061	347	695
2022	3295	6142	4417	3302	2010	1260	725	582
2023	5498	16912	7483	2540	1859	825	675	814
2024	6317	9053	14807	3950	1128	749	396	584

Table 14.3.7. Saithe in subareas 4 and 6 and in Division 3.a. Discards numbers (thousands) at age for the age range used in the assessment.

Year/Age	3	4	5	6	7	8	9	10+
1967	9617	3175	1141	55	16	7	5	2
1968	12888	4156	969	143	4	6	3	2
1969	16779	3461	813	181	57	19	16	2
1970	20671	15007	1102	288	38	42	8	3
1971	38741	11313	2406	112	42	67	48	9
1972	26680	12938	1861	504	36	69	42	18
1973	30152	7380	1973	381	83	47	35	26
1974	37148	6605	1038	491	144	122	38	22
1975	31626	5063	758	180	135	220	89	30
1976	115333	10387	860	148	39	126	92	38
1977	15240	10760	1084	130	37	81	97	52
1978	19456	5338	1327	79	11	24	20	38
1979	9063	2783	822	207	28	17	14	23
1980	9709	2416	662	160	56	27	13	25
1981	14527	1632	495	87	33	80	17	28
1982	17700	7945	674	142	21	34	25	14
1983	15673	4033	1912	135	48	35	26	15
1984	44915	6298	952	309	16	29	9	7
1985	74378	10885	976	113	30	16	10	8
1986	30764	17857	1116	128	26	34	11	10
1987	17326	19080	1028	75	18	29	23	8
1988	18079	5169	2596	91	19	26	20	7
1989	12280	6020	969	260	15	17	11	5
1990	22647	3833	832	112	27	18	7	4
1991	25593	5847	549	85	17	22	9	5
1992	10213	6580	937	76	15	16	13	5
1993	20990	4077	871	74	15	36	21	11
1994	11076	7868	958	102	9	8	11	9

Year/Age	3	4	5	6	7	8	9	10+
1995	14797	5096	1087	90	34	16	14	9
1996	6141	7607	866	184	31	19	5	5
1997	8344	3778	2218	87	26	24	7	3
1998	5751	6072	1203	382	22	28	14	3
1999	5233	2716	1961	199	99	55	23	7
2000	3920	3386	657	293	31	76	16	5
2001	8908	3454	1648	80	35	42	41	3
2002	8439	5710	2451	425	64	324	121	96
2003	15288	7106	234	0	0	0	0	0
2004	5605	4407	0	0	0	0	0	0
2005	3498	0	0	0	0	0	0	0
2006	5114	5282	394	0	0	0	0	0
2007	9433	3152	1762	97	0	0	0	0
2008	696	7682	1610	745	111	9	0	0
2009	831	1158	395	30	93	16	14	11
2010	886	390	266	117	1	11	0	38
2011	2636	1470	129	44	7	25	1	8
2012	7305	1341	1377	58	7	1	4	1
2013	2268	4801	339	224	4	0	0	1
2014	955	2205	1816	220	77	4	0	1
2015	2163	931	704	232	17	3	0	2
2016	3874	3867	905	573	26	7	1	0
2017	1943	3850	978	69	2	0	0	2
2018	586	3256	1922	333	17	11	2	1
2019	785	445	1016	597	39	11	1	1
2020	514	953	383	110	133	8	4	8
2021	436	163	51	36	0	0	0	0
2022	1493	1559	389	197	36	2	0	0
2023	876	1782	321	30	4	1	0	0

Year/Age	3	4	5	6	7	8	9	10+
2024	1179	1217	1585	113	0	0	0	0

Table 14.3.8. Saithe in subareas 4 and 6 and in Division 3.a. Catch weight-at-age (kg).

Year/Age	3	4	5	6	7	8	9	10+
1967	0.898	1.339	2.094	3.183	3.753	5.316	5.891	7.719
1968	1.234	1.624	1.979	3.007	4.039	4.428	6.136	7.406
1969	0.933	1.530	2.251	2.711	3.558	4.406	5.220	6.767
1970	0.908	1.416	2.049	2.716	3.599	4.463	5.687	6.845
1971	0.811	1.325	2.167	2.934	3.765	4.634	5.172	6.163
1972	0.780	1.175	1.952	2.367	3.793	4.228	4.630	6.326
1973	0.792	1.382	1.633	2.569	3.356	4.684	4.814	6.445
1974	0.831	1.534	2.372	2.751	3.428	4.498	5.713	7.857
1975	0.862	1.472	2.479	3.298	3.764	4.296	5.540	7.562
1976	0.678	1.287	2.250	3.068	4.034	4.383	5.112	7.147
1977	0.733	1.234	1.926	3.108	4.161	4.605	4.859	6.542
1978	0.793	1.304	2.145	3.338	4.521	4.900	5.449	7.400
1979	1.069	1.595	2.228	3.093	4.049	5.274	6.308	7.955
1980	0.921	1.790	2.380	3.028	4.089	5.126	5.939	8.148
1981	0.927	1.790	2.705	3.584	4.535	5.478	6.980	8.724
1982	1.048	1.548	2.518	3.218	4.206	5.125	5.905	8.823
1983	0.992	1.688	2.139	3.135	3.690	4.632	5.505	8.453
1984	0.767	1.586	2.286	2.688	3.895	4.665	6.183	8.474
1985	0.640	1.244	1.941	2.769	3.406	4.950	5.865	8.854
1986	0.670	1.018	1.786	2.430	3.571	4.209	5.651	8.218
1987	0.650	0.861	1.815	3.072	4.209	5.330	6.128	8.603
1988	0.752	0.964	1.379	2.789	4.023	5.254	6.322	8.649
1989	0.864	1.018	1.413	1.997	3.913	5.017	6.430	8.431
1990	0.815	1.175	1.575	2.245	3.241	4.858	6.315	8.416
1991	0.764	1.138	1.744	2.363	3.165	4.222	6.066	8.191

Year/Age	3	4	5	6	7	8	9	10+
1992	0.930	1.169	1.599	2.240	3.667	4.330	5.412	7.045
1993	0.868	1.239	1.746	2.634	3.184	3.980	5.080	6.891
1994	0.911	1.100	1.594	2.432	3.617	4.787	6.548	8.326
1995	0.967	1.272	1.807	2.560	3.554	4.767	5.267	7.891
1996	0.933	1.167	1.798	2.366	2.951	4.705	6.092	8.382
1997	0.873	1.125	1.445	2.585	3.555	4.525	6.158	8.866
1998	0.861	0.949	1.386	1.743	2.948	3.883	4.996	7.227
1999	0.850	1.042	1.206	1.752	2.337	3.493	4.844	6.745
2000	0.992	1.107	1.532	1.683	2.593	3.084	4.773	7.461
2001	0.774	1.053	1.307	2.093	2.546	3.485	4.141	6.141
2002	0.776	1.014	1.495	1.791	2.961	3.761	4.638	5.750
2003	0.636	0.889	1.167	1.810	2.368	3.176	3.768	5.065
2004	0.794	1.010	1.392	1.896	2.860	3.687	4.814	7.059
2005	0.715	1.155	1.325	1.710	2.132	3.026	3.622	5.713
2006	0.904	1.012	1.489	1.906	2.424	3.058	4.318	5.734
2007	0.769	1.124	1.286	1.834	2.328	2.887	3.600	4.975
2008	0.916	1.065	1.488	1.692	2.210	2.792	3.206	4.565
2009	1.033	1.333	1.672	1.994	2.566	3.086	3.651	4.790
2010	1.037	1.474	2.033	2.597	3.163	3.488	3.968	5.223
2011	0.955	1.192	1.787	2.571	3.068	3.418	3.718	4.289
2012	0.910	1.287	1.383	2.196	3.221	3.536	4.181	4.482
2013	0.878	1.132	1.586	1.957	3.076	3.841	4.541	5.648
2014	1.091	1.265	1.568	2.334	2.607	4.010	5.530	6.679
2015	0.951	1.253	1.621	2.180	3.037	3.793	4.228	7.285
2016	0.937	1.239	1.611	2.231	2.888	3.450	4.331	6.208
2017	0.956	1.228	1.755	2.356	2.987	4.232	4.473	6.287
2018	1.095	1.239	1.549	2.234	3.112	3.867	4.465	6.708
2019	1.133	1.442	1.809	2.320	3.081	3.897	4.677	6.613
2020	1.061	1.529	1.914	2.439	3.106	4.038	4.918	6.985

Year/Age	3	4	5	6	7	8	9	10+
2021	1.043	1.413	1.899	2.365	2.984	4.038	5.179	6.852
2022	1.079	1.459	1.918	2.541	3.262	3.960	4.797	7.130
2023	0.787	1.170	1.642	2.288	2.901	3.589	4.094	5.311
2024	0.853	1.125	1.423	1.952	2.851	3.601	4.350	5.935

Table 14.3.9. Saithe in subareas 4 and 6 and in Division 3.a. Landings weight-at-age (kg).

Year/Age	3	4	5	6	7	8	9	10+
1967	0.931	1.362	2.104	3.186	3.754	5.316	5.891	7.719
1968	1.278	1.652	1.989	3.009	4.040	4.428	6.136	7.406
1969	0.966	1.557	2.261	2.713	3.559	4.406	5.220	6.768
1970	0.941	1.441	2.059	2.718	3.600	4.463	5.687	6.845
1971	0.840	1.348	2.178	2.936	3.766	4.634	5.173	6.163
1972	0.808	1.196	1.961	2.369	3.794	4.228	4.630	6.326
1973	0.821	1.406	1.641	2.571	3.357	4.684	4.814	6.445
1974	0.861	1.561	2.383	2.753	3.429	4.498	5.713	7.857
1975	0.893	1.498	2.490	3.300	3.765	4.296	5.540	7.562
1976	0.702	1.309	2.260	3.071	4.035	4.383	5.112	7.147
1977	0.760	1.256	1.935	3.111	4.162	4.605	4.859	6.542
1978	0.822	1.327	2.155	3.340	4.522	4.901	5.449	7.400
1979	1.107	1.623	2.238	3.095	4.050	5.274	6.308	7.955
1980	0.955	1.821	2.391	3.030	4.090	5.126	5.939	8.148
1981	0.961	1.821	2.718	3.587	4.536	5.478	6.980	8.724
1982	1.086	1.575	2.529	3.220	4.207	5.125	5.905	8.823
1983	1.028	1.718	2.149	3.138	3.691	4.632	5.505	8.453
1984	0.795	1.614	2.297	2.690	3.896	4.665	6.183	8.474
1985	0.663	1.265	1.951	2.772	3.407	4.950	5.865	8.854
1986	0.694	1.035	1.794	2.432	3.572	4.209	5.651	8.218
1987	0.674	0.876	1.824	3.075	4.210	5.330	6.128	8.603
1988	0.779	0.981	1.386	2.791	4.024	5.254	6.322	8.649
1989	0.895	1.036	1.420	1.998	3.914	5.018	6.430	8.431

Year/Age	3	4	5	6	7	8	9	10+
1990	0.844	1.196	1.583	2.247	3.242	4.858	6.315	8.416
1991	0.791	1.158	1.752	2.365	3.165	4.222	6.066	8.191
1992	0.964	1.189	1.607	2.242	3.668	4.330	5.413	7.046
1993	0.899	1.260	1.754	2.636	3.185	3.980	5.080	6.891
1994	0.944	1.119	1.601	2.434	3.618	4.787	6.548	8.326
1995	1.002	1.294	1.816	2.562	3.555	4.767	5.267	7.891
1996	0.967	1.187	1.807	2.368	2.952	4.705	6.092	8.382
1997	0.905	1.145	1.452	2.587	3.556	4.525	6.158	8.866
1998	0.892	0.966	1.393	1.744	2.949	3.883	4.996	7.227
1999	0.881	1.061	1.211	1.754	2.337	3.493	4.844	6.745
2000	1.027	1.127	1.539	1.684	2.594	3.084	4.773	7.462
2001	0.802	1.072	1.313	2.095	2.546	3.485	4.141	6.141
2002	0.923	1.035	1.478	1.769	2.947	3.426	4.407	5.674
2003	0.833	0.980	1.173	1.810	2.368	3.176	3.768	5.065
2004	0.918	1.084	1.392	1.896	2.860	3.687	4.814	7.059
2005	0.921	1.155	1.325	1.710	2.132	3.026	3.622	5.713
2006	0.945	1.069	1.514	1.906	2.424	3.058	4.318	5.734
2007	0.837	1.143	1.317	1.840	2.328	2.887	3.600	4.975
2008	0.944	1.193	1.565	1.720	2.226	2.795	3.206	4.565
2009	1.036	1.340	1.664	1.992	2.563	3.085	3.648	4.793
2010	1.036	1.479	2.034	2.597	3.164	3.488	3.968	5.199
2011	1.007	1.207	1.783	2.573	3.068	3.404	3.717	4.284
2012	1.015	1.321	1.408	2.201	3.223	3.536	4.177	4.482
2013	0.898	1.156	1.614	1.976	3.078	3.841	4.541	5.648
2014	1.126	1.300	1.607	2.384	2.617	4.013	5.530	6.679
2015	0.977	1.244	1.625	2.190	3.043	3.796	4.228	7.287
2016	0.998	1.292	1.628	2.283	2.892	3.453	4.333	6.208
2017	1.047	1.302	1.809	2.361	2.988	4.232	4.473	6.292
2018	1.153	1.287	1.575	2.266	3.107	3.868	4.463	6.707

Year/Age	3	4	5	6	7	8	9	10+
2019	1.147	1.448	1.829	2.343	3.094	3.905	4.680	6.616
2020	1.066	1.542	1.938	2.447	3.132	4.043	4.912	6.984
2021	1.044	1.416	1.903	2.362	2.984	4.038	5.179	6.852
2022	1.160	1.530	1.955	2.574	3.277	3.959	4.797	7.130
2023	0.771	1.177	1.651	2.299	2.903	3.590	4.095	5.311
2024	0.856	1.139	1.451	1.966	2.851	3.601	4.350	5.935

Table 14.3.10. Saithe in subareas 4 and 6 and in Division 3.a. Discards weight-at-age (kg). Landing weight-at-age is reported and used when zero discards (Table 14.3.7).

Year/Age	3	4	5	6	7	8	9	10+
1967	0.748	1.076	1.818	2.972	3.590	5.316	5.891	7.719
1968	1.028	1.306	1.719	2.808	3.864	4.428	6.136	7.406
1969	0.777	1.230	1.955	2.531	3.403	4.406	5.220	6.767
1970	0.757	1.139	1.780	2.536	3.442	4.463	5.687	6.845
1971	0.676	1.065	1.882	2.739	3.601	4.634	5.172	6.163
1972	0.650	0.945	1.695	2.210	3.628	4.228	4.630	6.326
1973	0.660	1.111	1.419	2.399	3.210	4.684	4.814	6.445
1974	0.692	1.233	2.060	2.568	3.279	4.498	5.713	7.857
1975	0.718	1.184	2.153	3.079	3.600	4.296	5.540	7.562
1976	0.565	1.035	1.954	2.865	3.858	4.383	5.112	7.147
1977	0.611	0.993	1.673	2.902	3.980	4.605	4.859	6.542
1978	0.661	1.049	1.862	3.116	4.325	4.900	5.449	7.400
1979	0.890	1.283	1.935	2.888	3.873	5.274	6.308	7.955
1980	0.768	1.439	2.067	2.827	3.911	5.126	5.939	8.148
1981	0.773	1.439	2.349	3.346	4.338	5.478	6.980	8.724
1982	0.873	1.245	2.186	3.004	4.023	5.125	5.905	8.823
1983	0.826	1.358	1.858	2.927	3.529	4.632	5.505	8.453
1984	0.639	1.276	1.985	2.510	3.726	4.665	6.183	8.474
1985	0.533	1.000	1.686	2.586	3.258	4.950	5.865	8.854
1986	0.558	0.818	1.551	2.269	3.416	4.209	5.651	8.218
1987	0.542	0.693	1.576	2.869	4.026	5.330	6.128	8.603

Year/Age	3	4	5	6	7	8	9	10+
1988	0.626	0.775	1.198	2.604	3.848	5.254	6.322	8.649
1989	0.720	0.819	1.227	1.865	3.743	5.017	6.430	8.431
1990	0.679	0.945	1.368	2.097	3.100	4.858	6.315	8.416
1991	0.636	0.915	1.515	2.206	3.027	4.222	6.066	8.191
1992	0.775	0.940	1.389	2.092	3.508	4.330	5.412	7.045
1993	0.723	0.996	1.517	2.460	3.046	3.980	5.080	6.891
1994	0.759	0.884	1.384	2.271	3.459	4.787	6.548	8.326
1995	0.806	1.023	1.570	2.390	3.400	4.767	5.267	7.891
1996	0.778	0.938	1.562	2.209	2.823	4.705	6.092	8.382
1997	0.728	0.905	1.255	2.413	3.400	4.525	6.158	8.866
1998	0.717	0.764	1.204	1.627	2.820	3.883	4.996	7.227
1999	0.708	0.838	1.047	1.636	2.235	3.493	4.844	6.745
2000	0.826	0.890	1.330	1.571	2.480	3.084	4.773	7.461
2001	0.645	0.847	1.135	1.955	2.435	3.485	4.141	6.141
2002	0.616	0.896	1.580	2.483	3.469	6.058	6.935	6.927
2003	0.469	0.571	0.641	1.689	2.265	3.176	3.768	5.065
2004	0.617	0.676	1.203	1.769	2.735	3.687	4.814	7.059
2005	0.741	0.913	1.146	1.595	2.038	3.026	3.622	5.713
2006	0.808	0.724	0.859	1.778	2.318	3.058	4.318	5.734
2007	0.660	1.062	0.949	1.365	2.227	2.887	3.600	4.975
2008	0.633	0.680	0.967	1.161	1.495	1.820	3.206	2.797
2009	1.010	1.253	1.946	2.403	2.838	3.388	3.934	3.911
2010	1.046	1.374	1.987	2.561	3.025	3.351	3.968	6.895
2011	0.756	0.971	2.054	2.445	3.170	4.072	4.369	6.618
2012	0.808	0.997	1.101	1.831	2.675	3.411	4.804	5.313
2013	0.835	1.003	1.180	1.300	2.298	3.841	4.541	5.861
2014	0.977	1.072	1.274	1.487	2.077	3.223	5.530	7.568
2015	0.877	1.326	1.531	1.848	2.410	2.184	4.228	5.911
2016	0.882	1.096	1.440	1.764	2.384	2.864	2.634	4.282

Year/Age	3	4	5	6	7	8	9	10+
2017	0.815	0.937	1.269	1.907	2.484	4.232	4.473	2.817
2018	0.894	1.049	1.318	1.554	3.770	3.715	5.371	7.697
2019	1.033	1.336	1.537	1.932	2.162	2.991	2.816	2.969
2020	1.042	1.379	1.456	1.937	2.306	3.448	5.480	7.101
2021	1.025	1.305	1.466	2.583	2.984	4.038	5.179	6.852
2022	0.900	1.177	1.496	1.988	2.474	4.033	4.797	7.130
2023	0.881	1.100	1.419	1.415	2.218	2.876	3.753	3.486
2024	0.838	1.021	1.156	1.487	2.470	3.601	4.350	5.935

Table 14.3.11. Saithe in subareas 4 and 6 and in Division 3.a. Stock weight-at-age (kg).

Year/Age	3	4	5	6	7	8	9	10+
1967	0.604	0.974	1.659	2.743	3.493	5.275	6.033	8.239
1968	0.829	1.182	1.569	2.591	3.760	4.394	6.283	7.905
1969	0.627	1.114	1.784	2.336	3.312	4.372	5.346	7.224
1970	0.611	1.031	1.624	2.340	3.350	4.429	5.824	7.307
1971	0.545	0.964	1.718	2.528	3.504	4.598	5.297	6.578
1972	0.524	0.856	1.547	2.039	3.531	4.195	4.742	6.753
1973	0.533	1.006	1.295	2.213	3.124	4.648	4.930	6.879
1974	0.558	1.117	1.880	2.370	3.191	4.463	5.851	8.387
1975	0.579	1.072	1.965	2.841	3.503	4.263	5.673	8.072
1976	0.456	0.937	1.783	2.644	3.755	4.350	5.235	7.629
1977	0.493	0.899	1.526	2.678	3.873	4.569	4.976	6.983
1978	0.533	0.949	1.700	2.876	4.208	4.863	5.581	7.899
1979	0.718	1.161	1.766	2.665	3.769	5.234	6.460	8.491
1980	0.619	1.303	1.886	2.609	3.806	5.087	6.082	8.697
1981	0.623	1.303	2.144	3.088	4.221	5.436	7.149	9.312
1982	0.704	1.127	1.995	2.772	3.915	5.086	6.047	9.418
1983	0.667	1.229	1.696	2.701	3.434	4.596	5.638	9.023
1984	0.516	1.155	1.812	2.316	3.625	4.629	6.332	9.045
1985	0.430	0.905	1.539	2.386	3.170	4.912	6.006	9.451

Year/Age	3	4	5	6	7	8	9	10+
1986	0.450	0.741	1.416	2.093	3.324	4.177	5.787	8.772
1987	0.437	0.627	1.439	2.647	3.918	5.289	6.276	9.183
1988	0.505	0.702	1.093	2.403	3.744	5.214	6.475	9.232
1989	0.581	0.741	1.120	1.721	3.642	4.979	6.585	8.999
1990	0.547	0.856	1.249	1.935	3.017	4.821	6.467	8.984
1991	0.513	0.828	1.382	2.036	2.946	4.190	6.212	8.744
1992	0.625	0.851	1.267	1.930	3.413	4.296	5.543	7.520
1993	0.583	0.902	1.384	2.270	2.964	3.949	5.203	7.355
1994	0.612	0.801	1.263	2.095	3.366	4.750	6.706	8.887
1995	0.650	0.926	1.433	2.206	3.308	4.730	5.394	8.423
1996	0.627	0.850	1.425	2.039	2.747	4.669	6.239	8.947
1997	0.587	0.819	1.146	2.227	3.309	4.490	6.306	9.464
1998	0.578	0.691	1.099	1.502	2.744	3.853	5.116	7.715
1999	0.571	0.759	0.955	1.510	2.175	3.467	4.961	7.200
2000	0.666	0.806	1.214	1.450	2.414	3.061	4.888	7.964
2001	0.520	0.767	1.036	1.804	2.369	3.458	4.241	6.555
2002	0.521	0.738	1.185	1.543	2.756	3.732	4.750	6.138
2003	0.554	0.767	1.056	1.746	2.440	3.805	4.616	7.571
2004	0.537	0.712	0.963	1.377	2.260	2.933	4.221	5.227
2005	0.556	0.776	1.066	1.412	2.011	3.092	3.719	5.986
2006	0.663	0.772	1.155	1.599	2.137	2.948	3.931	5.710
2007	0.585	0.902	1.049	1.643	2.136	2.907	3.587	5.829
2008	0.674	0.884	1.373	1.629	2.533	2.946	3.687	5.133
2009	0.800	1.080	1.421	2.123	2.466	3.529	3.893	5.803
2010	0.776	1.205	1.717	2.266	3.252	3.611	4.741	6.053
2011	0.525	0.854	1.486	2.124	2.763	3.636	3.891	5.374
2012	0.489	0.816	1.101	2.106	2.943	3.718	4.400	5.179
2013	0.606	0.766	1.176	1.581	2.987	3.891	4.676	6.073
2014	0.677	0.894	1.195	1.812	2.407	4.225	5.039	6.523

Year/Age	3	4	5	6	7	8	9	10+
2015	0.612	0.950	1.316	1.861	2.580	3.382	5.330	7.108
2016	0.552	0.853	1.262	1.845	2.616	3.532	4.178	6.698
2017	0.567	0.784	1.275	2.067	2.859	3.689	4.617	6.791
2018	0.645	0.821	1.144	1.834	2.789	3.591	4.234	6.155
2019	0.748	1.001	1.298	1.821	2.680	3.718	4.487	6.269
2020	0.756	1.158	1.480	2.020	2.868	3.790	4.827	7.180
2021	0.661	1.077	1.592	2.143	2.848	3.852	4.615	6.449
2022	0.589	0.879	1.383	2.065	2.782	3.496	4.594	6.940
2023	0.573	0.840	1.180	2.030	2.990	3.792	4.528	6.494
2024	0.523	0.697	0.976	1.526	2.586	3.513	4.160	5.362

Table 14.3.12. Saithe in subareas 4 and 6 and in Division 3.a. Proportion mature-at-age.

Year/Age	3	4	5	6	7	8	9	10+
1967	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1968	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1969	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1970	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1971	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1972	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1973	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1974	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1975	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1976	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1977	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1978	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1979	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1980	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1981	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1982	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965

Year/Age	3	4	5	6	7	8	9	10+
1983	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1984	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1985	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1986	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1987	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1988	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1989	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1990	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1991	0.1888	0.3445	0.5662	0.8061	0.9410	0.9804	0.9920	0.9965
1992	0.1785	0.3587	0.5832	0.8217	0.9416	0.9811	0.9920	0.9965
1993	0.2110	0.3669	0.5843	0.8174	0.9435	0.9808	0.9927	0.9962
1994	0.1891	0.3434	0.5492	0.8107	0.9423	0.9817	0.9916	0.9964
1995	0.1816	0.3296	0.5690	0.8114	0.9446	0.9801	0.9916	0.9969
1996	0.1788	0.3291	0.5616	0.8023	0.9368	0.9803	0.9926	0.9971
1997	0.1969	0.3564	0.5705	0.8083	0.9449	0.9804	0.9928	0.9971
1998	0.1996	0.3478	0.5670	0.8036	0.9398	0.9803	0.9919	0.9967
1999	0.1758	0.3416	0.5555	0.7913	0.9377	0.9799	0.9916	0.9960
2000	0.1880	0.3271	0.5551	0.7881	0.9375	0.9787	0.9914	0.9960
2001	0.2123	0.3397	0.5472	0.8143	0.9435	0.9803	0.9917	0.9964
2002	0.2504	0.3767	0.5504	0.7646	0.9354	0.9809	0.9878	0.9962
2003	0.2437	0.4263	0.5954	0.8011	0.9338	0.9829	0.9917	0.9968
2004	0.2158	0.4159	0.6503	0.8252	0.9369	0.9812	0.9894	0.9961
2005	0.2074	0.3384	0.6063	0.8339	0.9396	0.9805	0.9906	0.9968
2006	0.2596	0.3633	0.5411	0.8239	0.9542	0.9834	0.9914	0.9959
2007	0.2639	0.4182	0.5696	0.7852	0.9502	0.9861	0.9919	0.9959
2008	0.2582	0.4781	0.6721	0.8310	0.9390	0.9867	0.9945	0.9966
2009	0.2226	0.3556	0.6508	0.8479	0.9465	0.9801	0.9926	0.9970
2010	0.1096	0.2911	0.5045	0.8287	0.9460	0.9774	0.9893	0.9968
2011	0.2155	0.2760	0.5704	0.8027	0.9570	0.9862	0.9920	0.9965

Year/Age	3	4	5	6	7	8	9	10+
2012	0.1721	0.3061	0.4519	0.7976	0.9322	0.9838	0.9931	0.9956
2013	0.1776	0.2849	0.4711	0.7162	0.9351	0.9777	0.9934	0.9965
2014	0.1817	0.3316	0.4897	0.7577	0.9191	0.9826	0.9921	0.9970
2015	0.1932	0.3629	0.5709	0.7668	0.9234	0.9737	0.9926	0.9969
2016	0.1400	0.2938	0.5283	0.7769	0.9086	0.9691	0.9864	0.9957
2017	0.1536	0.2403	0.4620	0.7582	0.9176	0.9612	0.9852	0.9946
2018	0.2701	0.3062	0.4676	0.7710	0.9351	0.9785	0.9880	0.9945
2019	0.2614	0.4113	0.5264	0.7442	0.9291	0.9787	0.9916	0.9944
2020	0.1709	0.3830	0.5893	0.7571	0.9037	0.9753	0.9901	0.9950
2021	0.1501	0.3263	0.6106	0.8279	0.9225	0.9704	0.9888	0.9954
2022	0.1362	0.2528	0.5210	0.8173	0.9404	0.9730	0.9843	0.9957
2023	0.1525	0.2731	0.4708	0.8010	0.9510	0.9835	0.9893	0.9943
2024	0.2147	0.2581	0.4431	0.6897	0.9179	0.9824	0.9913	0.9943

Table 14.3.13. Saithe in subareas 4 and 6 and in Division 3.a. Data available for calibration of the final assessment. Indices include one commercial standardized CPUE index (year effects), tuned to the exploitable biomass within SAM, and indices for age 3–8 from combined research surveys in quarters 3 and 4 (delta-GAM approach for divisions 3.a and 6.a, and Subarea 4).

Year	Survey index–Q3–4 (delta-GAM) at age						CPUE
	3	4	5	6	7	8	
1992	1.3430	0.9411	0.1875	0.0479	0.0463	0.0519	
1993	3.9985	0.7771	0.3219	0.0768	0.0198	0.0349	
1994	1.1357	0.6999	0.1777	0.1352	0.0284	0.0139	
1995	3.7091	0.8686	0.3204	0.0784	0.0712	0.0323	
1996	1.8997	2.2984	0.2145	0.1360	0.0434	0.0112	
1997	2.0540	1.0460	1.4469	0.1461	0.1203	0.0466	
1998	1.3402	2.3966	0.6308	0.6791	0.0790	0.0825	
1999	1.4772	0.6937	1.0582	0.1785	0.1600	0.0368	
2000	2.0726	3.8705	0.4731	0.3283	0.0632	0.0665	1.0000
2001	8.5159	2.6783	1.6472	0.1928	0.2086	0.0537	1.2522
2002	3.4928	4.5350	0.6181	0.4903	0.0826	0.1075	1.0117

Year	Survey index–Q3–4 (delta-GAM) at age						CPUE
	3	4	5	6	7	8	
2003	5.4167	4.2884	1.8158	0.1651	0.1366	0.0641	0.9129
2004	2.3158	2.0319	1.3140	0.4619	0.0541	0.0510	1.0938
2005	4.0668	2.0930	1.2732	0.5172	0.2621	0.0392	1.1783
2006	2.6732	5.0529	0.9367	0.4289	0.1933	0.1127	1.1451
2007	4.6510	0.9188	1.6217	0.2901	0.1718	0.0847	1.0548
2008	1.8215	1.6485	0.3084	0.6181	0.1460	0.1245	1.1054
2009	1.3428	0.8695	0.3315	0.0664	0.1044	0.0670	0.9536
2010	3.9073	1.0816	0.4408	0.2150	0.0400	0.1281	0.8977
2011	4.4644	3.3135	0.5933	0.3802	0.1245	0.0254	0.9924
2012	5.3127	1.4554	1.4713	0.2874	0.2143	0.1249	0.9560
2013	3.9795	4.4195	0.6317	0.6542	0.1646	0.0778	0.9973
2014	1.8891	1.6931	1.1762	0.2162	0.2553	0.0565	0.8423
2015	3.6925	2.1391	2.1303	0.6766	0.1351	0.1493	1.0160
2016	5.9564	3.9889	1.3606	1.0254	0.3690	0.1021	0.9489
2017	5.3635	6.1222	1.6198	0.4702	0.3212	0.1574	1.0408
2018	1.0648	1.9918	1.8248	0.4691	0.1237	0.0583	1.0798
2019	0.5778	0.7411	0.8527	0.3530	0.0972	0.0361	0.9007
2020	0.8560	1.4928	0.8194	0.4876	0.2669	0.0754	0.8206
2021	1.2117	0.5058	0.3994	0.1954	0.1195	0.0636	0.7294
2022	1.7324	1.0638	0.4831	0.2524	0.1597	0.1136	0.8623
2023	1.2411	3.1531	0.5994	0.1408	0.0785	0.0427	0.8659
2024	1.8017	1.4943	1.4551	0.2450	0.0396	0.0227	0.7381

Table 14.4.1. Saithe in subareas 4 and 6 and in Division 3.a. Model configuration for the SAM assessment.

```
# Configuration saved: Tue Mar 12 15:05:46 2024
#
# Where a matrix is specified, rows corresponds to fleets and columns to ages.
# Same number indicates same parameter used
# Numbers (integers) starts from zero and must be consecutive
# Negative numbers indicate that the parameter is not included in the model
#
$minAge
# The minimium age class in the assessment
 3

$maxAge
# The maximum age class in the assessment
 10

$maxAgePlusGroup
# Is last age group considered a plus group for each fleet (1 yes, or 0 no).
 1 0 1

$keyLogFsta
# Coupling of the fishing mortality states processes for each age (normally
only
# the first row (= fleet) is used).
# Sequential numbers indicate that the fishing mortality is estimated indi-
vidually
# for those ages; if the same number is used for two or more ages, F is bound
for
# those ages (assumed to be the same). Binding fully selected ages will result
in a
# flat selection pattern for those ages.
   0   1   2   3   4   5   6   6
  -1  -1  -1  -1  -1  -1  -1  -1
  -1  -1  -1  -1  -1  -1  -1  -1

$corFlag
# Correlation of fishing mortality across ages (0 independent, 1 compound
symmetry,
# 2 AR(1), 3 separable AR(1).
# 0: independent means there is no correlation between F across age
# 1: compound symmetry means that all ages are equally correlated;
# 2: AR(1) first order autoregressive - similar ages are more highly correlated
than
# ages that are further apart, so similar ages have similar F patterns over
time.
# if the estimated correlation is high, then the F pattern over time for each
age
# varies in a similar way. E.g if almost one, then they are parallel (like a
```

```
# separable model) and if almost zero then they are independent.
# 3: Separable AR - Included for historic reasons . . .  more later
 2

$keyLogFpar
# Coupling of the survey catchability parameters (nomally first row is
# not used, as that is covered by fishing mortality).
  -1  -1  -1  -1  -1  -1  -1  -1
   0   1   2   3   4   5  -1  -1
   6  -1  -1  -1  -1  -1  -1  -1

$keyQpow
# Density dependent catchability power parameters (if any).
  -1  -1  -1  -1  -1  -1  -1  -1
  -1  -1  -1  -1  -1  -1  -1  -1
  -1  -1  -1  -1  -1  -1  -1  -1

$keyVarF
# Coupling of process variance parameters for log(F)-process (Fishing mortality
# normally applies to the first (fishing) fleet; therefore only first row is
used)
   0   1   2   3   3   3   3   3
  -1  -1  -1  -1  -1  -1  -1  -1
  -1  -1  -1  -1  -1  -1  -1  -1

$keyVarLogN
# Coupling of the recruitment and survival process variance parameters for the
# log(N)-process at the different ages. It is advisable to have at least the
first age
# class (recruitment) separate, because recruitment is a different process
than
# survival.
 0 1 1 1 1 1 1 1

$keyVarObs
# Coupling of the variance parameters for the observations.
# First row refers to the coupling of the variance parameters for the catch
data
# observations by age
# Second and further rows refers to coupling of the variance parameters for
the
# index data observations by age
   0   1   1   2   2   2   2   2
   3   3   3   3   3   3  -1  -1
   4  -1  -1  -1  -1  -1  -1  -1

$obsCorStruct
# Covariance structure for each fleet ("ID" independent, "AR" AR(1), or "US"
for unstructured). | Possible values are: "ID" "AR" "US"
```

```
 "AR" "AR" "ID"

$keyCorObs
# Coupling of correlation parameters can only be specified if the AR(1) struc-
ture is chosen above.
# NA's indicate where correlation parameters can be specified (-1 where they
cannot).
#V1 V2 V3 V4 V5 V6 V7
   0   0   0   1   1   1   1
   2   2   2   2   3  -1  -1
  NA  -1  -1  -1  -1  -1  -1

$stockRecruitmentModelCode
# Stock recruitment code (0 for plain random walk, 1 for Ricker, 2 for Beverton-
Holt,
# 3 piece-wise constant, 61 for segmented regression/hockey stick, 62 for
AR(1),
# 63 for bent hyperbola / smooth hockey stick, 64 for power function with
degree < 1,
# 65 for power function with degree > 1, 66 for Shepher, 67 for Deriso,
# 68 for Saila-Lorda, 69 for sigmoidal Beverton-Holt, 90 for CMP spline,
# 91 for more flexible spline, and 92 for most flexible spline).
 0

$noScaledYears
# Number of years where catch scaling is applied.
 0

$fbarRange
# lowest and higest age included in Fbar
 4 7

$keyBiomassTreat
# To be defined only if a biomass survey is used (0 SSB index, 1 catch index,
# 2 FSB index, 3 total catch, 4 total landings, 5 TSB index, 6 TSN index, and
10 Fbar idx).
 -1 -1 2

$obsLikelihoodFlag
# Option for observational likelihood | Possible values are: "LN" "ALN"
 "LN" "LN" "LN"

$fixVarToWeight
# If weight attribute is supplied for observations this option sets the treat-
ment
# (0 relative weight, 1 fix variance to weight). Can be specified fleetwise.
 0

$fracMixF
```

```
# The fraction of t(3) distribution used in logF increment distribution
 0

$fracMixN
# The fraction of t(3) distribution used in logN increment distribution
# (for each age group)
 0

$fracMixObs
# A vector with same length as number of fleets, where each element is the
fraction
# of t(3) distribution used in the distribution of that fleet
 0 0 0

$predVarObsLink
# Coupling of parameters used in a prediction-variance link for observations.
  -1  -1  -1  -1  -1  -1  -1  -1
  -1  -1  -1  -1  -1  -1  NA  NA
  NA  NA  NA  NA  NA  NA  NA  NA

$stockWeightModel
# Integer code describing the treatment of stock weights in the model (0 use
as known,
# 1 use as observations to inform stock weight process (GMRF with cohort and
# within year correlations)), 2 to add extra correlation to plusgroup
 0

$keyStockWeightMean
# Coupling of stock-weight process mean parameters (not used if stockWeight-
Model==0)
 NA NA NA NA NA NA NA NA

$keyStockWeightObsVar
# Coupling of stock-weight observation variance parameters
# (not used if stockWeightModel==0)
 NA NA NA NA NA NA NA NA

$catchWeightModel
# Integer code describing the treatment of catch weights in the model (0 use
as known,
# 1 use as observations to inform catch weight process (GMRF with cohort and
within
# year correlations)), 2 to add extra correlation to plusgroup
 0

$keyCatchWeightMean
# Coupling of catch-weight process mean parameters (not used if catchWeight-
Model==0)
  NA  NA  NA  NA  NA  NA  NA  NA
```

```
$keyCatchWeightObsVar
# Coupling of catch-weight observation variance parameters
# (not used if catchWeightModel==0)
  NA  NA  NA  NA  NA  NA  NA  NA

$matureModel
# Integer code describing the treatment of proportion mature in the model (0
use as
# known, 1 use as observations to inform proportion mature process (GMRF with
cohort
# and within year correlations on logit(proportion mature))),
# 2 to add extra correlation to plusgroup
 0

$keyMatureMean
# Coupling of mature process mean parameters (not used if matureModel==0)
 NA NA NA NA NA NA NA NA

$mortalityModel
# Integer code describing the treatment of natural mortality in the model (0
use as
# known, 1 use as observations to inform natural mortality process (GMRF with
cohort
# and within year correlations)), 2 to add extra correlation to plusgroup
 0

$keyMortalityMean
#
 NA NA NA NA NA NA NA NA

$keyMortalityObsVar
# Coupling of natural mortality observation variance parameters
# (not used if mortalityModel==0)
 NA NA NA NA NA NA NA NA

$keyXtraSd
# An integer matrix with 4 columns (fleet year age coupling),
# which allows additional uncertainty to be estimated for the specified ob-
servations

$logNMeanAssumption
#
 0 0

$initState
#
 0
```

Table 14.4.2. Saithe in subareas 4 and 6 and in Division 3.a. Fishing mortalities at age for the final assessment model. F at age 9 and 10+ are coupled in the model (9+).

Year/Age	3	4	5	6	7	8	9+
1967	0.190	0.279	0.299	0.369	0.345	0.327	0.347
1968	0.205	0.302	0.309	0.372	0.348	0.329	0.349
1969	0.199	0.294	0.301	0.370	0.347	0.329	0.347
1970	0.270	0.394	0.354	0.391	0.365	0.345	0.360
1971	0.317	0.437	0.370	0.399	0.374	0.354	0.367
1972	0.371	0.482	0.388	0.407	0.383	0.363	0.373
1973	0.444	0.550	0.421	0.421	0.396	0.376	0.383
1974	0.518	0.624	0.463	0.437	0.413	0.390	0.394
1975	0.508	0.615	0.476	0.444	0.421	0.399	0.400
1976	0.603	0.729	0.544	0.469	0.443	0.418	0.416
1977	0.499	0.623	0.519	0.464	0.441	0.418	0.414
1978	0.435	0.539	0.490	0.458	0.437	0.415	0.410
1979	0.351	0.441	0.453	0.449	0.431	0.410	0.405
1980	0.322	0.413	0.452	0.452	0.435	0.415	0.409
1981	0.270	0.359	0.427	0.446	0.432	0.413	0.408
1982	0.347	0.478	0.513	0.478	0.458	0.436	0.427
1983	0.374	0.544	0.560	0.495	0.472	0.447	0.437
1984	0.457	0.697	0.642	0.519	0.490	0.462	0.450
1985	0.532	0.878	0.733	0.543	0.509	0.477	0.465
1986	0.527	0.959	0.786	0.557	0.520	0.486	0.475
1987	0.453	0.826	0.739	0.546	0.511	0.478	0.470
1988	0.400	0.725	0.702	0.537	0.503	0.471	0.465
1989	0.398	0.722	0.700	0.536	0.501	0.468	0.463
1990	0.391	0.714	0.698	0.535	0.499	0.465	0.461
1991	0.382	0.723	0.710	0.538	0.502	0.466	0.463
1992	0.358	0.720	0.729	0.543	0.507	0.470	0.468
1993	0.308	0.651	0.694	0.535	0.505	0.470	0.470
1994	0.297	0.668	0.717	0.542	0.513	0.477	0.478

Year/Age	3	4	5	6	7	8	9+
1995	0.233	0.550	0.648	0.525	0.502	0.470	0.473
1996	0.174	0.416	0.563	0.502	0.483	0.455	0.459
1997	0.151	0.364	0.518	0.488	0.471	0.446	0.452
1998	0.148	0.356	0.512	0.487	0.469	0.447	0.452
1999	0.134	0.331	0.487	0.480	0.462	0.443	0.448
2000	0.113	0.276	0.437	0.463	0.446	0.428	0.435
2001	0.130	0.319	0.462	0.472	0.452	0.433	0.438
2002	0.125	0.311	0.450	0.470	0.452	0.433	0.438
2003	0.120	0.294	0.411	0.457	0.441	0.424	0.429
2004	0.105	0.262	0.370	0.439	0.426	0.411	0.415
2005	0.103	0.262	0.368	0.437	0.424	0.408	0.408
2006	0.112	0.288	0.383	0.441	0.429	0.411	0.408
2007	0.118	0.321	0.409	0.448	0.435	0.416	0.409
2008	0.128	0.378	0.467	0.467	0.453	0.431	0.420
2009	0.111	0.341	0.446	0.459	0.446	0.426	0.413
2010	0.092	0.295	0.411	0.445	0.435	0.416	0.402
2011	0.101	0.328	0.432	0.450	0.439	0.419	0.403
2012	0.094	0.317	0.414	0.443	0.431	0.411	0.395
2013	0.080	0.277	0.379	0.429	0.419	0.400	0.383
2014	0.064	0.227	0.343	0.415	0.407	0.389	0.372
2015	0.062	0.216	0.337	0.412	0.404	0.384	0.368
2016	0.063	0.225	0.348	0.417	0.409	0.387	0.369
2017	0.060	0.218	0.338	0.414	0.407	0.383	0.364
2018	0.069	0.244	0.358	0.422	0.414	0.387	0.366
2019	0.078	0.265	0.369	0.426	0.418	0.387	0.365
2020	0.082	0.274	0.365	0.421	0.413	0.379	0.357
2021	0.075	0.239	0.329	0.404	0.397	0.363	0.342
2022	0.068	0.211	0.308	0.393	0.386	0.352	0.332
2023	0.076	0.226	0.320	0.398	0.389	0.353	0.332

Year/Age	3	4	5	6	7	8	9+
2024	0.081	0.235	0.326	0.400	0.391	0.354	0.332

Table 14.4.3. Saithe in subareas 4 and 6 and in Division 3.a. Estimated population numbers-at-age for the final assessment model (thousands).

Year/Age	3	4	5	6	7	8	9	10+
1967	197413	97971	71804	6976	4667	1021	630	570
1968	212875	111407	57712	38388	3691	2299	653	650
1969	422722	105681	58385	29643	19233	2462	1756	598
1970	391303	286082	54725	37087	16462	9747	1612	1360
1971	442807	219998	134239	24071	18319	10063	5925	2067
1972	276949	234011	104539	68233	12794	9887	5714	5161
1973	241678	123196	104241	62566	32653	7713	5003	7114
1974	235427	99359	47702	57824	39165	16604	4717	6766
1975	306677	84674	35944	22196	32735	22270	9241	7038
1976	451435	128392	32078	17522	12066	17195	11590	10004
1977	199479	165271	42064	12773	8470	6962	9211	12239
1978	145336	88184	61049	16664	5541	4493	3346	12729
1979	110769	59351	37888	28401	8282	2794	2314	9041
1980	107049	52319	27967	18336	15279	3571	1579	7145
1981	228344	45757	26394	12020	9059	7922	1694	5635
1982	172638	139271	25191	15469	5975	4657	3527	4007
1983	193656	75446	68283	10840	8367	2948	2420	3595
1984	312883	89266	33652	28972	4826	3964	1407	2951
1985	451312	126095	32580	14773	10776	2527	1687	2560
1986	416418	176643	33696	12788	7061	4603	1277	2325
1987	187584	220707	35982	10674	5660	3565	2335	1789
1988	176931	77929	77130	11065	4925	2773	1891	1865
1989	129710	81968	31167	26288	4796	2343	1386	1718
1990	181496	54952	28636	13064	9299	2587	1133	1511
1991	201440	79519	19237	11314	6040	3961	1343	1478

Year/Age	3	4	5	6	7	8	9	10+
1992	134138	91938	26310	7799	5296	2966	2083	1538
1993	234994	61674	31311	8200	3335	2959	1628	2100
1994	165064	107955	26856	13830	3377	1546	1472	2148
1995	281639	79298	38755	9980	6482	1417	991	1740
1996	160214	171766	30423	16956	6411	2330	646	1401
1997	207673	89106	94275	12633	8348	3381	1049	997
1998	117885	143970	45572	46191	6770	3904	1813	965
1999	166499	63497	83783	19932	20966	4139	1803	1403
2000	141040	120466	30581	36877	10558	11131	1990	1429
2001	287545	87055	76115	13755	16299	6548	6681	1352
2002	190799	166203	37635	34311	6821	7769	3572	4217
2003	205467	132980	78752	14504	14661	4106	4154	4089
2004	151446	119257	71060	40475	6711	6586	3080	3521
2005	205303	85042	68850	41726	20607	4213	3447	3543
2006	115838	150202	42172	33197	21260	10205	2604	3757
2007	210832	54831	85134	23118	17344	12078	5160	3742
2008	110163	110553	30060	46572	13400	8624	7634	6099
2009	87377	63413	43557	14078	21148	7899	4427	8296
2010	148783	46619	31227	19107	6584	10972	4504	7883
2011	131442	101211	26303	16812	9438	3559	5002	9361
2012	211718	55943	58077	13569	8118	4735	1925	8675
2013	135717	134501	22853	29001	7323	4069	2348	6144
2014	93569	82697	59054	12572	14383	4049	1844	4972
2015	133324	54864	57682	29177	7097	6566	2629	3886
2016	165709	81258	33774	29352	12885	4715	3106	4090
2017	132782	116538	41496	16743	14301	6223	2753	3968
2018	64960	86978	69796	22363	7470	6665	3330	3896
2019	67669	44182	51386	29759	9099	3660	3361	3882
2020	47157	49572	27671	24126	14047	4036	1931	3523

Year/Age	3	4	5	6	7	8	9	10+
2021	86146	32509	23918	14730	12502	6173	1978	3262
2022	135392	52558	20877	13588	8246	6006	3540	2907
2023	84375	117465	31823	10476	7377	3942	3204	3630
2024	90081	55864	67057	16929	5028	3779	2204	3469

Table 14.5.1. Saithe in subareas 4 and 6 and in Division 3.a. Estimated recruitment, total-stock biomass (TSB, tonnes), spawning-stock biomass (SSB, tonnes), and average fishing mortality for ages 4 to 7 ($F_{4–7}$), 1967–2025. Low and High refer to the lower and upper 95% confidence interval estimates.

Year	$R_{(age\ 3)}$			SSB			$F_{4–7}$			TSB		
	Median	Low	High	Median	Low	High	Median	Low	High	Median	Low	High
1967	197413	138355	281682	167342	136672	204896	0.323	0.261	0.400	383088	310731	472294
1968	212875	151767	298587	242281	198122	296282	0.333	0.272	0.407	531422	434250	650339
1969	422722	301043	593584	289471	238178	351810	0.328	0.269	0.400	644217	525002	790503
1970	391303	281430	544071	380408	316771	456830	0.376	0.313	0.451	827123	687776	994701
1971	442807	320985	610863	448718	374637	537447	0.395	0.331	0.471	900323	755786	1072500
1972	276949	201776	380127	444906	369799	535268	0.415	0.350	0.493	794871	671717	940606
1973	241678	176121	331638	459413	378849	557109	0.447	0.378	0.529	737557	623263	872809
1974	235427	170868	324379	498445	408547	608125	0.484	0.411	0.571	752523	634747	892153
1975	306677	223962	419942	465211	381020	568004	0.489	0.418	0.573	720851	609109	853092
1976	451435	322864	631207	402185	332430	486575	0.546	0.464	0.642	686536	580648	811735
1977	199479	144779	274844	326335	270663	393458	0.512	0.438	0.598	541150	461567	634454
1978	145336	105539	200139	302912	251214	365250	0.481	0.413	0.560	477208	405877	561076
1979	110769	80279	152840	272678	226382	328443	0.444	0.380	0.517	428589	363640	505140
1980	107049	77238	148364	248403	206203	299240	0.438	0.376	0.510	383097	324893	451727
1981	228344	164906	316186	251870	210581	301255	0.416	0.355	0.487	441505	372742	522954

Year	$R_{(age\ 3)}$			SSB			F_{4-7}			TSB		
	Median	Low	High	Median	Low	High	Median	Low	High	Median	Low	High
1982	172638	125674	237151	244035	207123	287527	0.482	0.417	0.557	477777	407005	560855
1983	193656	141648	264759	231655	197852	271233	0.518	0.447	0.599	455231	390106	531229
1984	312883	228322	428763	224456	192028	262361	0.587	0.508	0.679	463888	396181	543166
1985	451312	328242	620525	211254	182003	245205	0.666	0.573	0.774	474571	402285	559846
1986	416418	298574	580775	197662	170693	228892	0.705	0.598	0.832	463337	391285	548657
1987	187584	136629	257542	185498	158979	216442	0.655	0.566	0.759	372507	321116	432122
1988	176931	129569	241605	165725	141325	194339	0.617	0.534	0.712	317337	272722	369252
1989	129710	94777	177519	143715	121867	169481	0.615	0.533	0.709	269959	231797	314404
1990	181496	132599	248424	135001	114953	158545	0.611	0.531	0.704	268842	229149	315411
1991	201440	147880	274398	130011	111573	151496	0.618	0.538	0.710	274552	234152	321922
1992	134138	102532	175487	127352	109937	147525	0.625	0.544	0.718	264443	228261	306360
1993	234994	180527	305895	134451	116274	155470	0.597	0.521	0.684	300134	255760	352207
1994	165064	124531	218789	137632	118529	159814	0.610	0.527	0.706	298036	255203	348059
1995	281639	211801	374505	153628	132030	178760	0.556	0.482	0.641	382155	320224	456063
1996	160214	119985	213930	161729	138699	188584	0.491	0.428	0.564	369370	313682	434945
1997	207673	152788	282272	191329	163851	223415	0.460	0.401	0.527	389846	331110	459002

Year	$R_{(age\ 3)}$			SSB			F_{4-7}			TSB		
	Median	Low	High	Median	Low	High	Median	Low	High	Median	Low	High
1998	117885	87550	158729	181166	154497	212438	0.456	0.399	0.521	337451	288651	394500
1999	166499	124387	222870	177215	151832	206841	0.440	0.385	0.503	332446	285353	387312
2000	141040	104228	190854	190393	165048	219630	0.405	0.354	0.464	362348	313885	418294
2001	287545	218269	378809	213352	185370	245557	0.426	0.373	0.488	418498	361037	485105
2002	190799	145324	250504	224724	195167	258757	0.421	0.367	0.483	410334	355592	473503
2003	205467	156476	269797	239668	208121	275997	0.401	0.350	0.459	425873	368633	492002
2004	151446	115192	199110	207708	180647	238822	0.374	0.326	0.430	356244	309393	410191
2005	205303	154145	273440	225156	195449	259377	0.372	0.324	0.428	400845	347480	462404
2006	115838	87067	154117	236654	205706	272257	0.385	0.335	0.442	401860	350615	460595
2007	210832	161332	275520	243808	212542	279672	0.403	0.352	0.462	412486	359852	472818
2008	110163	83264	145752	272827	237168	313846	0.441	0.382	0.509	407975	357286	465855
2009	87377	65061	117347	247335	213592	286408	0.423	0.368	0.486	375611	328344	429682
2010	148783	111897	197828	219593	188081	256385	0.396	0.346	0.455	398624	345979	459279
2011	131442	99674	173334	196852	168818	229540	0.412	0.359	0.473	339061	295214	389422
2012	211718	162451	275925	176194	151404	205042	0.401	0.349	0.462	336531	291902	387985
2013	135717	103247	178398	173498	149523	201317	0.376	0.326	0.433	344065	298190	396998

Year	$R_{(age\ 3)}$			SSB			$F_{4–7}$			TSB		
	Median	**Low**	**High**	**Median**	**Low**	**High**	**Median**	**Low**	**High**	**Median**	**Low**	**High**
2014	93569	71028	123264	178021	153886	205942	0.348	0.302	0.401	324029	281306	373240
2015	133324	101604	174947	199612	173419	229763	0.342	0.297	0.394	346045	301030	397791
2016	165709	126650	216814	184618	159733	213381	0.350	0.304	0.403	348348	302130	401636
2017	132782	101720	173329	183111	158068	212122	0.344	0.299	0.397	357690	310957	411445
2018	64960	48547	86923	182816	158816	210443	0.360	0.312	0.414	316986	277390	362236
2019	67669	50156	91297	181968	158406	209035	0.369	0.321	0.426	293095	256475	334942
2020	47157	34960	63609	174835	151718	201474	0.368	0.318	0.426	272957	238986	311756
2021	86146	64541	114983	155244	133987	179874	0.342	0.295	0.397	251155	218481	288715
2022	135392	98648	185822	138612	118276	162444	0.324	0.277	0.380	263276	224912	308186
2023	84375	60254	118153	142514	121124	167682	0.333	0.283	0.392	280939	238773	330552
2024	90081	57416	141329	119541	100260	142531	0.338	0.280	0.407	231393	190223	281472
2025	94008*			135337**	105472**	171745**						

*** Recruitment in 2025 is the geometric mean of resampled recruitment estimates from 2015 to 2024.**
**** SSB 2025 is the STF, composed of 2024 survivors (age 4–10+) and stochastic recruitment and biological assumption 2025 (age 3; ≈ 8% of SSB)**

Table 14.7.1. Saithe in subareas 4 and 6 and in Division 3.a. Comparisons of forecast assumptions in the previous assessment (WGNSSK 2024) with estimates (and assumption for R 2025) in the current assessment. Assumptions or technical basis for catch and F given in brackets. Catches are given for age 3–10+.

Variable	Year*	Previous assessment (2024)	Current assessment (2025)
Recruitment** (thousands)	2024	93 961	90 81^
	2025	94 086	94 008
SSB (tonnes)	2024	185 632	120 700^
	2025	195 899	135 337
Catch (tonnes)	2024	78 251	60 303^^
$F_{ages\ 4–7}$	2024	0.320 (F_{sq})	0.338^
	2025	0.316 (F_{MSY})	0.338 (F_{sq})

*** ’2024’ = Intermediate year in the previous assessment; ‘2025’ = advice year in the previous assessment, intermediate year in the current assessment.**
^ Model estimate.
^^ InterCatch estimate.

Table 14.7.2. Saithe in subareas 4 and 6 and in Division 3.a. The basis for the catch options.

Variable	Value	Notes
$F_{ages\ 4–7}$ (2025)	0.338	Average exploitation pattern (2022–2024) rescaled to $F_{ages\ 4–7}$ (2024).
SSB (2026)	162 097	Short-term forecast (STF); tonnes
$R_{age\ 3}$ (2025)	93 865 (94 008)	Geometric mean of recruitment resampled from the years 2015–2024 (geometric mean of unique recruitment estimates, or geometric mean expectation, reported in the advice); thousands
$R_{age\ 3}$ (2026)	93 979 (94 008)	Geometric mean of recruitment resampled from the years 2015–2024 (geometric mean of unique recruitment estimates, or geometric mean expectation, reported in the advice); thousands
Total catch (2025)	62 705	STF, F *status quo* assumption; tonnes
Landings (2025)	59 243	STF, assuming 2022–2024 average landings fraction by age; tonnes
Discards (2025)	3462	STF, assuming 2022–2024 average discards fraction by age; tonnes

Table 14.7.3. Saithe in subareas 4 and 6 and in Division 3.a. Reference points and their technical basis.

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY $B_{trigger}$	180 770	B_{pa}; in tonnes.	ICES (2024)
	F_{MSY}	0.316	EqSim analysis based on the recruitment period 2002–2022, using a segmented regression with a breakpoint fixed at B_{lim}.	ICES (2024)
Precautionary approach	B_{lim}	130 090	B_{loss} (SSB in 1992); in tonnes.	ICES (2024)
	B_{pa}	180 770	$B_{lim} \times \exp(1.645 \times 0.2) \approx 1.4 \times B_{lim}$; in tonnes.	ICES (2024)
	F_{lim}	0.464	The F that on average leads to B_{lim} from EqSim (recruitment period 2002–2022).	ICES (2024)
	F_{pa}	0.392	The F that provides a 95% probability for SSB to be above B_{lim} ($F_{P.05}$ with advice rule [AR]).	ICES (2024)
EU Management plan*	MAP MSY $B_{trigger}$	180 770	MSY B_{trigge}; in tonnes.	ICES (2024)
	MAP B_{lim}	130 090	B_{lim}; in tonnes.	ICES (2024)
	MAP F_{MSY}	0.316	F_{MSY}	ICES (2024)
	MAP range F_{lower}	0.192–0.316	Consistent with ranges provided by ICES, resulting in no more than 5% reduction in long-term yield compared with MSY	ICES (2024)
	MAP range F_{upper}	0.316–0.392	Consistent with ranges provided by ICES, resulting in no more than 5% reduction in long-term yield compared with MSY. Capped at F_{pa}.	ICES (2024)

*** EU multiannual plan (MAP) for the North Sea (EU, 2018)**

Table 14.7.4. Saithe in subareas 4 and 6, and in Division 3.a. Annual catch scenarios. All weights are in tonnes.

Basis	Total catch (2026)	Projected landings (2026)	Projected discards^^ (2026)	Projected landings# 3.a4	Projected landings# 6	F_{total} (ages 4–7) (2026)	$F_{projected\ landings}$ (ages 4–7) (2026)	$F_{projected\ discards}$ (ages 4–7) (2026)	SSB (2027)	% SSB change *	% TAC change **	% advice change ^	Probability of SSB (2027) < B_{lim} (%)
ICES advice basis													
MSY approach: F_{MSY} × SSB (2026)/MSY $B_{trigger}$	60 167	56 835	3332	51 493	5342	0.283	0.27	0.0140	173 194	6.8	-24	-24	5.2
Other scenarios													
$F_{MSY\ lower}$ × SSB (2026)/MSY $B_{trigger}$	38 328	36 230	2098	32 824	3406	0.172	0.163	0.0090	190 394	17.5	-52	-52	1.18
$F_{MSY\ lower}$	42 383	40 059	2324	36 293	3766	0.192	0.182	0.0100	187 176	15.5	-46	-46	1.65
F_{MSY}	66 183	62 520	3663	56 643	5877	0.32	0.30	0.0170	168 502	4.0	-16.3	-16.3	7.4
F = 0	0	0	0	0	0	0	0	0	221 546	37	-100	-100	0.030
F_{pa}	79 535	75 081	4454	68 023	7058	0.392	0.37	0.021	158 004	-2.5	0.59	0.59	13.8
SSB (2027) = B_{lim}	117 051	110 255	6796	99 891	10 364	0.63	0.60	0.033	130 090	-19.7	48	48	50
SSB (2027) = B_{pa}= MSY $B_{trigger}$	51 240	48 422	2818	43 870	4552	0.24	0.22	0.0120	180 770	11.5	-35	-35	2.6
F = F_{2025}	70 101	66 228	3873	60 003	6225	0.34	0.32	0.0180	165 358	2.0	-11.3	-11.3	9.0
TAC_{2025}	79 071	74 647	4424	67 630	7017	0.39	0.37	0.020	158 394	-2.3	0	0	13.6
TAC_{2025} −15%	67 210	63 479	3731	57 512	5967	0.32	0.30	0.0170	167 681	3.4	-15.0	-15.0	7.9
TAC_{2025} +15%	90 932	85 756	5176	77 695	8061	0.46	0.44	0.024	149 193	-8.0	15.0	15.0	22
TAC_{2025} −20%	63 257	59 748	3509	54 132	5616	0.30	0.28	0.0160	170 759	5.3	-20	-20	6.3
TAC_{2025} +25%	98 839	93 194	5645	84 434	8760	0.51	0.48	0.027	143 134	-11.7	25	25	30

*** SSB_{2027} relative to SSB_{2026}.**

**** Total catch in 2026 relative to the TAC in 2025 (79 071 t).**

 # Projected landings split according to the average in 1993–1998, as agreed by the relevant management authorities, i.e. 90.6% in Subarea 4 and Subdivision 3.a.20 and 9.4% in Subarea

^ Total catch 2026 relative to the advice value 2025 (79 071 t).

^^ Including BMS landings. Assuming recent discard rate.

Table 14.7.5. Saithe in subareas 4 and 6 and in Division 3.a. Contribution of the year classes to the landings in 2026.

Year class	Age	Contribution to landings (%)	
		Numbers	Weight
2023	3	13	6.7
2022	4	26.5	18.5
2021	5	21.7	18.7
2020	6	14.8	15.4
2019	7	16.3	21.9
2018	8	3.8	7.2
2017	9	1.2	2.9
	10+	2.6	8.8

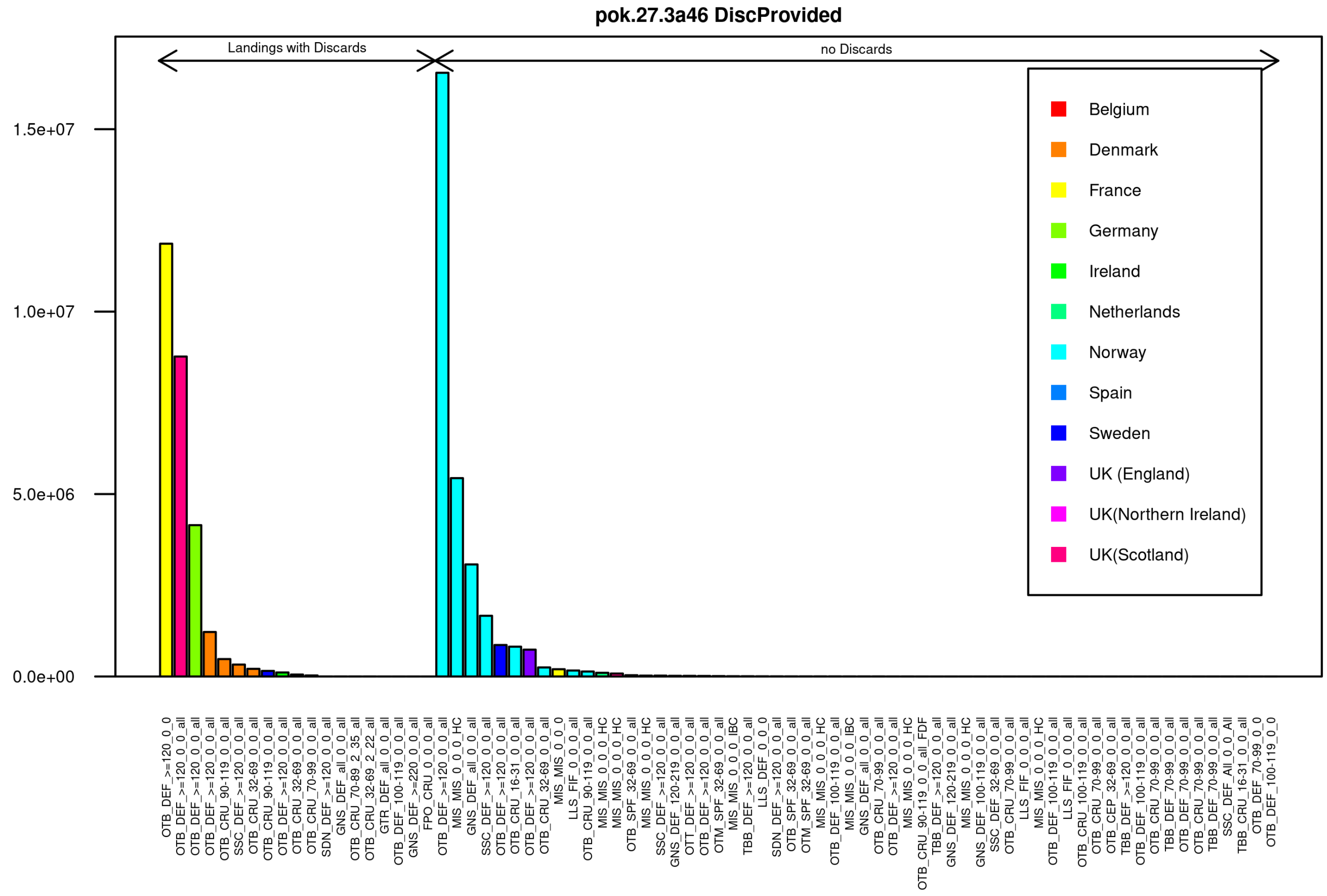

Figure 14.3.1. Saithe in subareas 4 and 6 and in Division 3.a. Landings with associated discards for areas and quarters combined by métier for 2024.

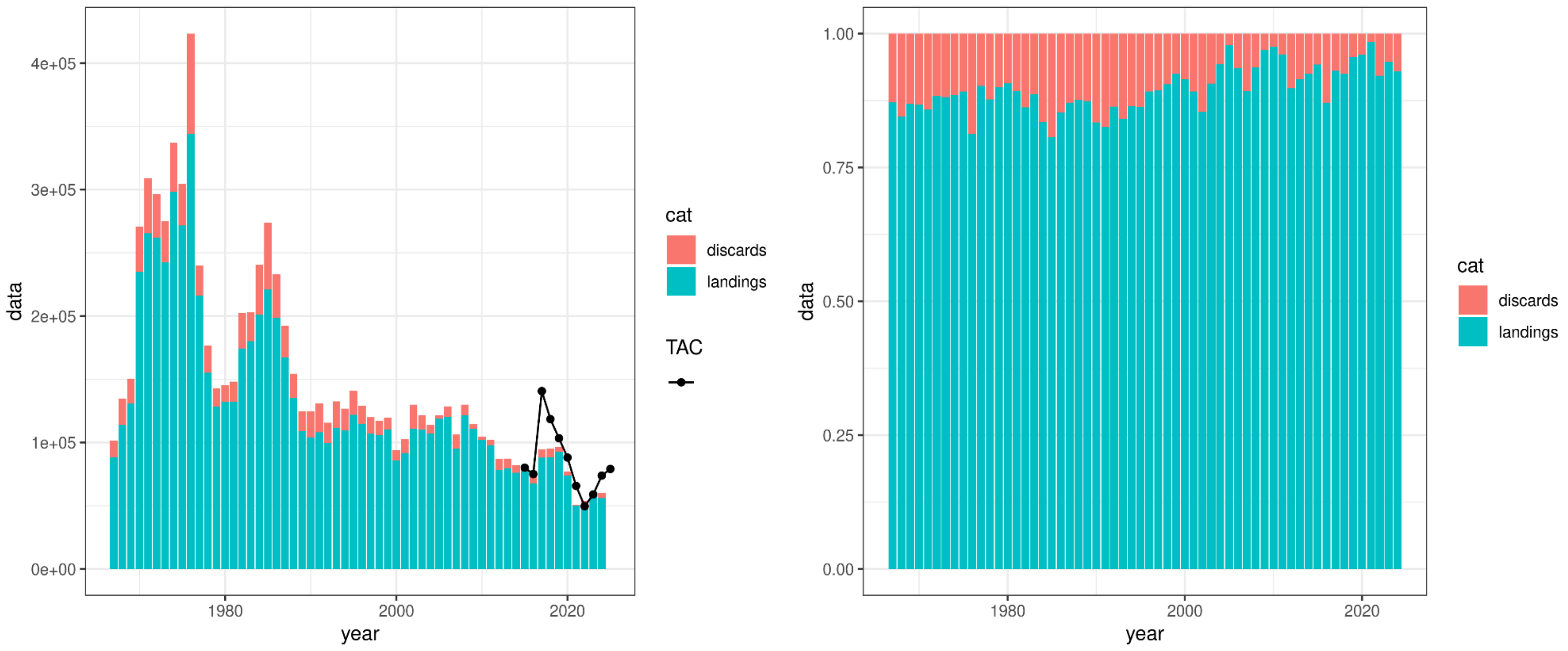

Figure 14.3.2. Saithe in subareas 4 and 6 and in Division 3.a. Yield as stacked plot for landings and discards in tonnes (left panel) and as percent (right panel). Landings include BMS landings from Norway since 2016. Discards correspond to unwanted catch (discards + EU/UK BMS) since 2016.

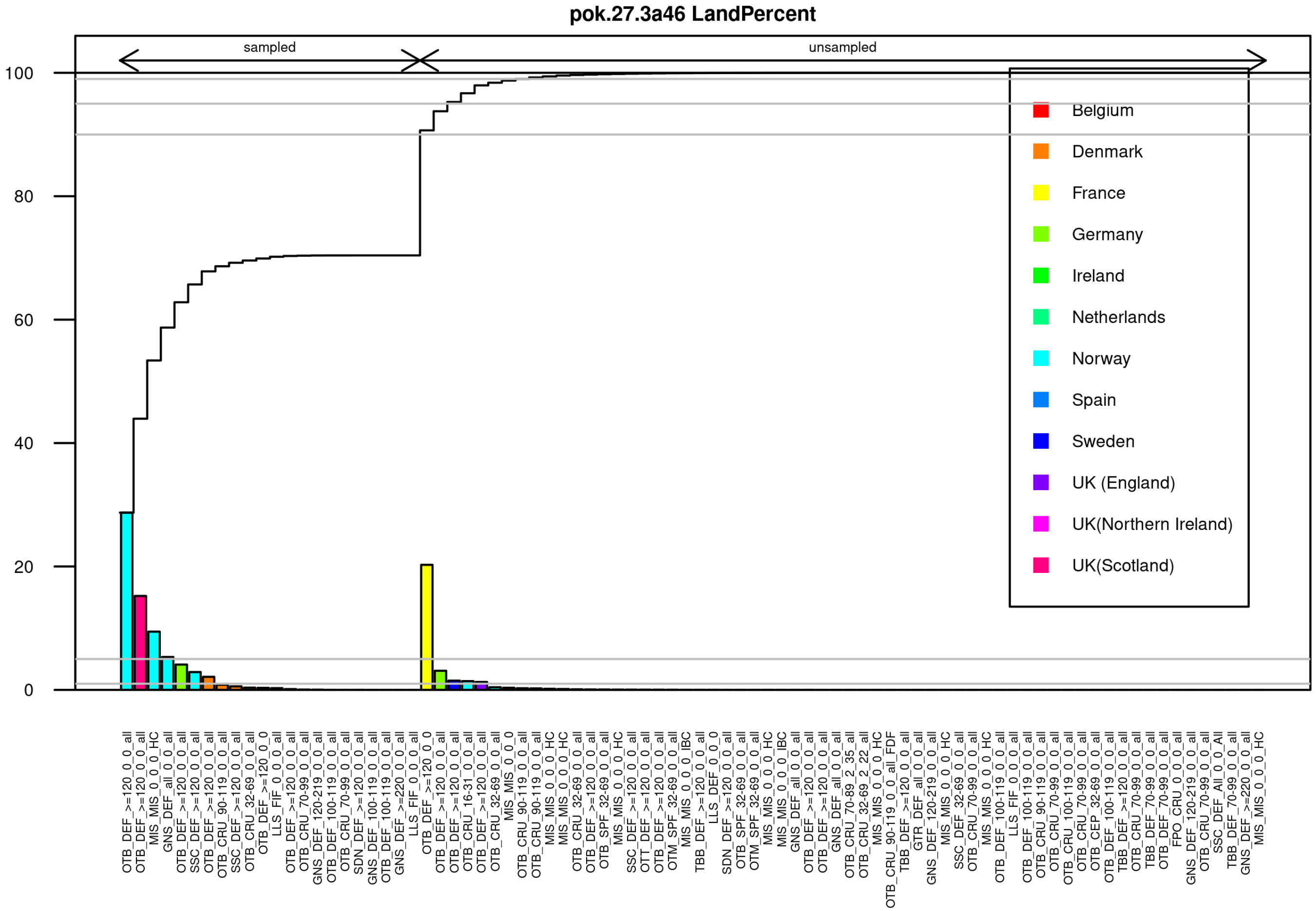

Figure 14.3.3. Saithe in subareas 4 and 6 and in Division 3.a. Overview of percent of sampled and unsampled landings by country and métier for 2024.

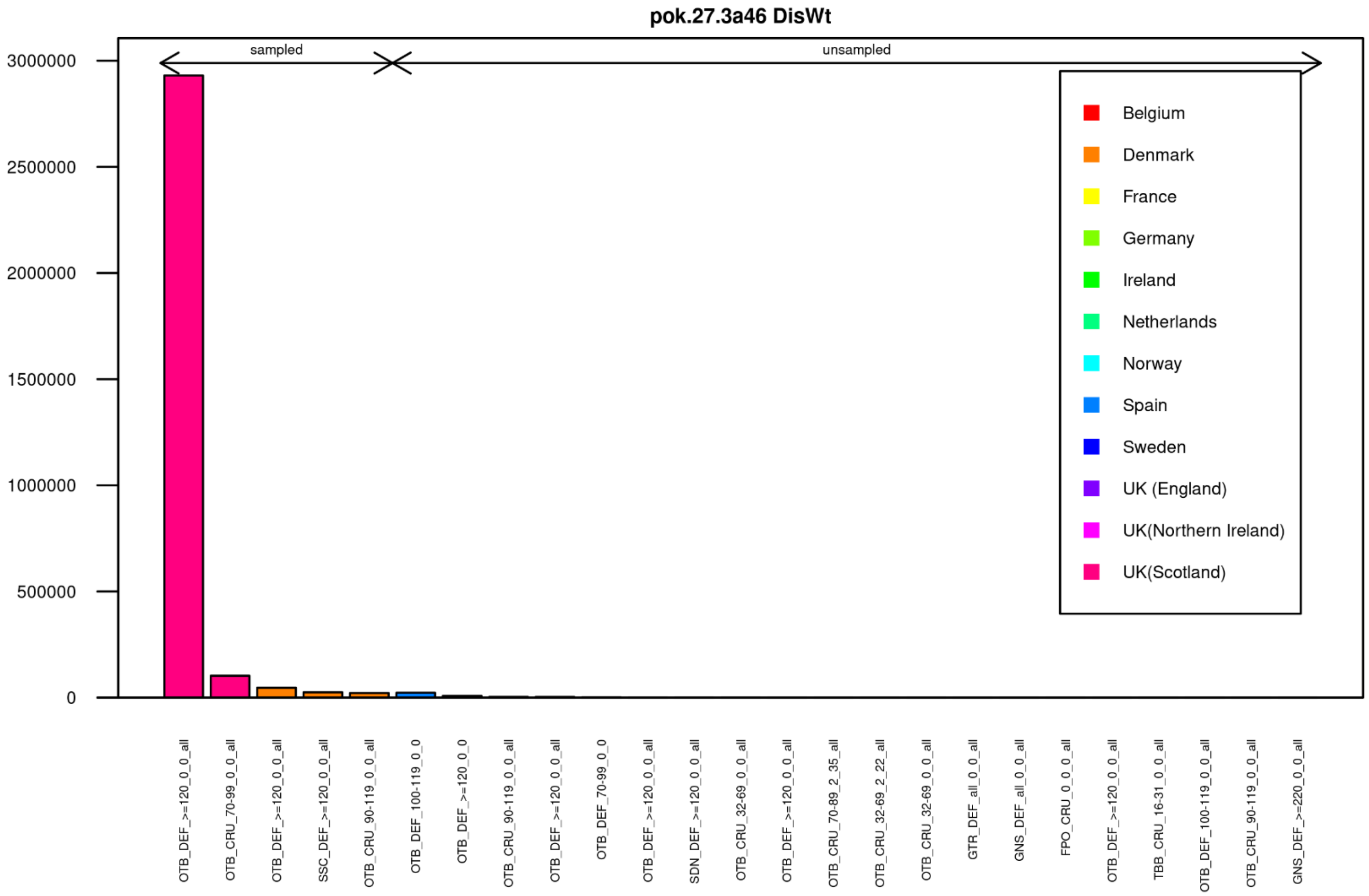

Figure 14.3.4. Saithe in subareas 4 and 6 and in Division 3.a. Overview of age sampled and unsampled imported discards by country and métier for 2024.

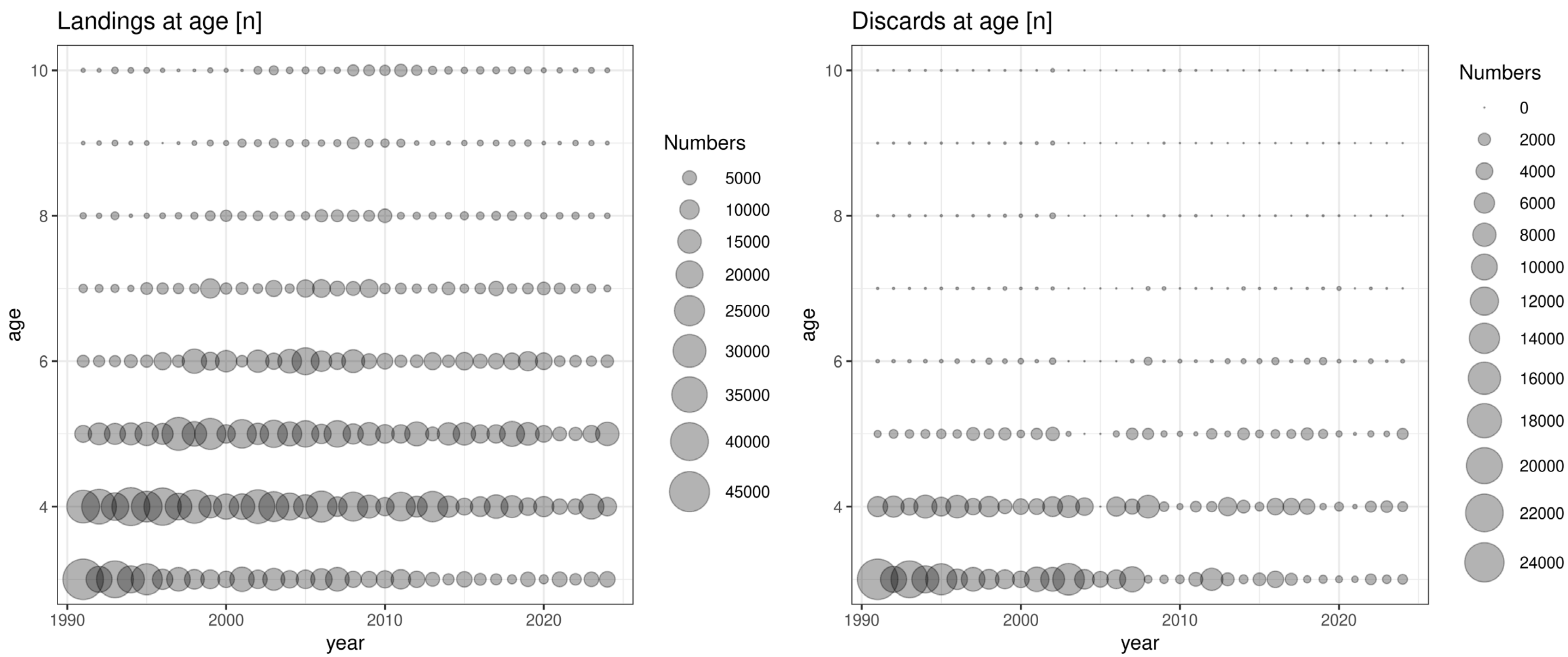

Figure 14.3.5. Saithe in subareas 4 and 6 and in Division 3.a. (left) Landings-at-age (x10³) for saithe ages 3–10+, 1990–2024. (Right) Discard numbers (x10³) at age for saithe ages 3–10+, 1990–2024.

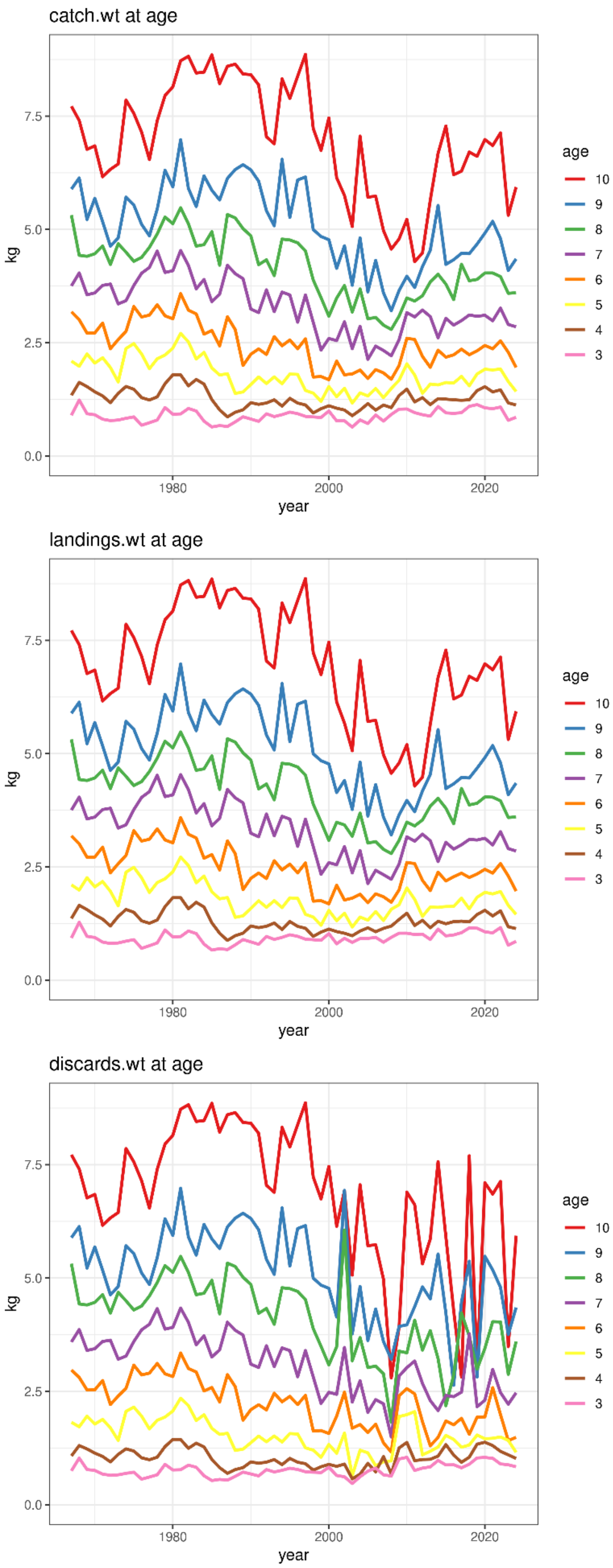

Figure 14.3.6. Saithe in subareas 4 and 6 and in Division 3.a. Catch weight-at-age (top panel), landing weight-at-age (middle panel) and discard weights-at-age (bottom panel), in kilograms, for saithe ages 3–10+, 1967–2024.

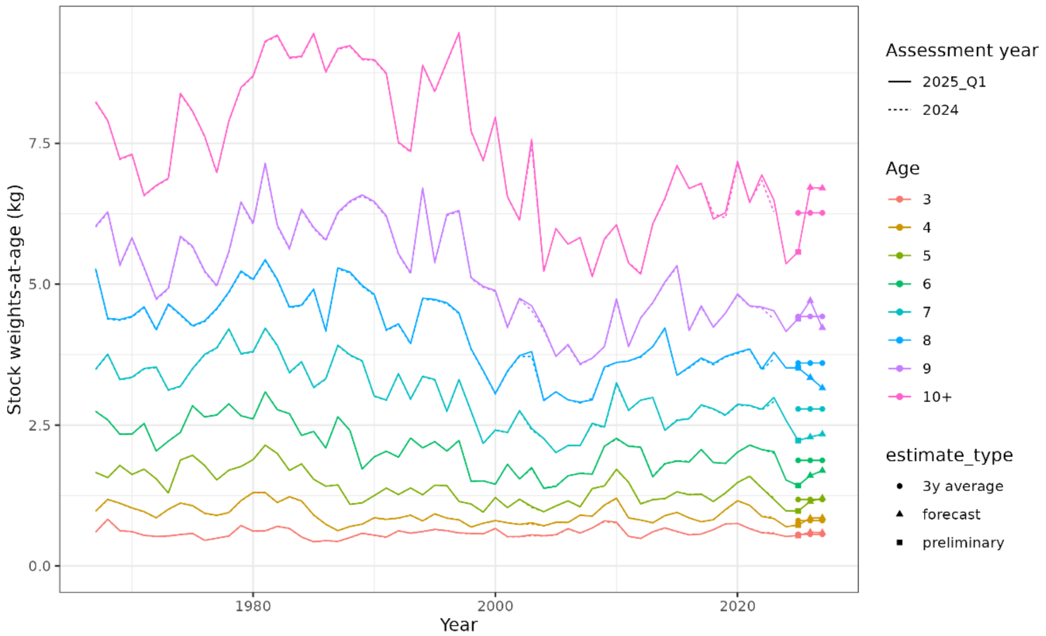

Figure 14.3.7. Saithe in subareas 4 and 6 and in Division 3.a. Model-based stock weights-at-age, for saithe ages 3–10+, 1967–2025 and, preliminary estimate (Q1 2025) and projections 2026-2027, as used in the forecast. Comparison with the previous assessment series (assessment year 2024) shows very little revisions of former years' estimates with the extra year of data. Preliminary estimates and projections are also compared to the last three-year average, formerly used in the forecast.

Figure 14.3.8. Saithe in subareas 4 and 6 and in Division 3.a. Model-based proportion mature-at-age, for saithe ages 3–10+, 1967–2024, and preliminary estimate (Q1 2025) and projections 2026-2027, as used in the forecast. Comparison with the previous assessment series (assessment year 2024) shows some revisions of former years' estimates with the two extra years of data. Preliminary estimates and projections are also compared to the last three-year average, formerly used in the forecast.

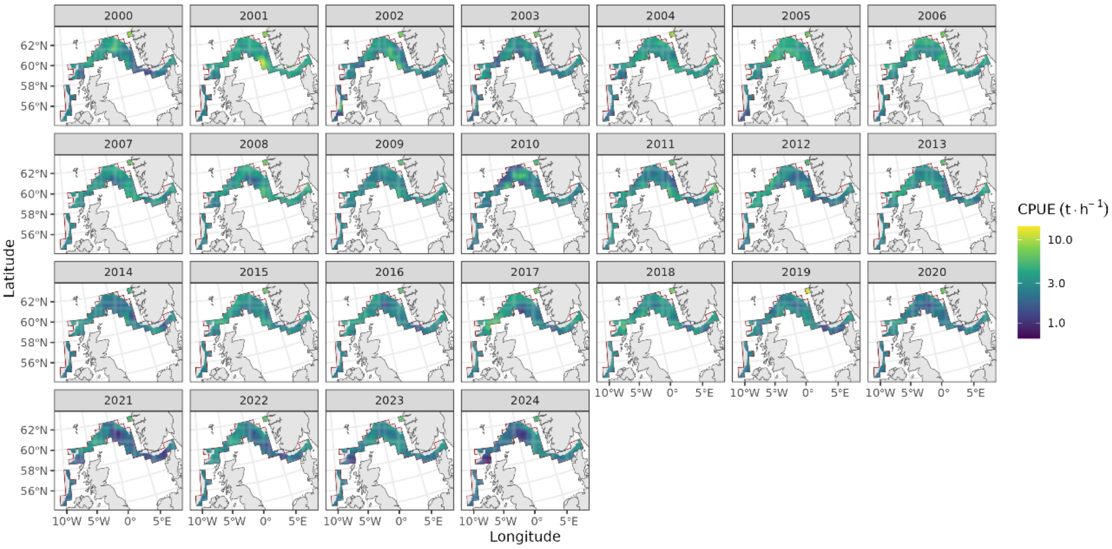

Figure 14.3.9. Saithe in subareas 4 and 6 and in Division 3.a. Yearly CPUE predictions on a fine grid (10x10 km cells), for a typical Norwegian vessel of power category > 6020 kW in January, as used in calculation of the area-weighted CPUE index (Figure 14.3.10).

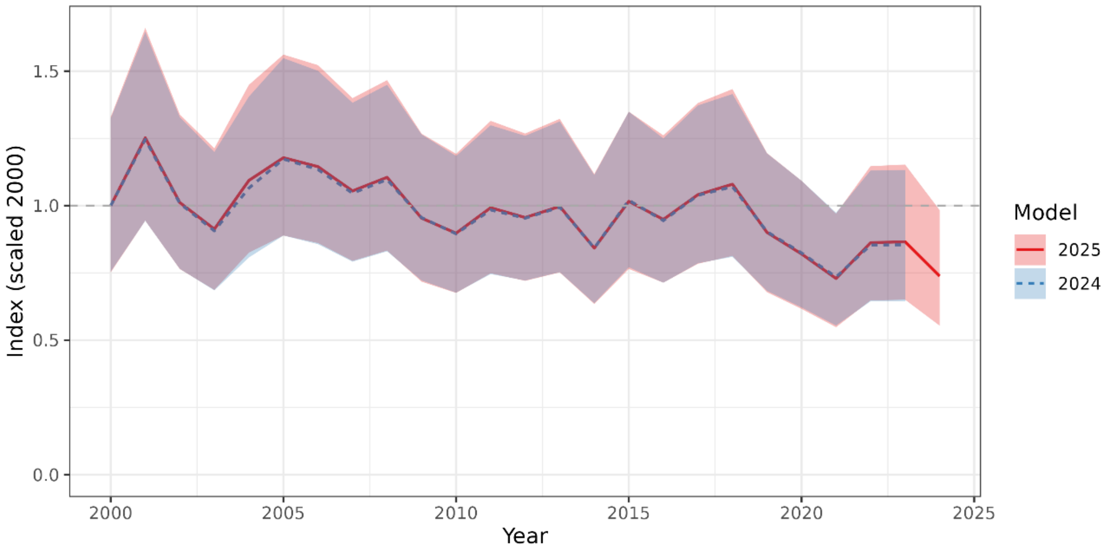

Figure 14.3.10. Saithe in subareas 4 and 6 and in Division 3.a. Scaled standardized commercial CPUE index time-series and 95% confidence interval, 2000–2024. Based on logbook data from France, Germany, and Norway. Comparison of the updated index (data 2024 and some corrections back in time; solid red line) to the previous 2000–2023 index, as used in the 2024 assessment (blue dashed line).

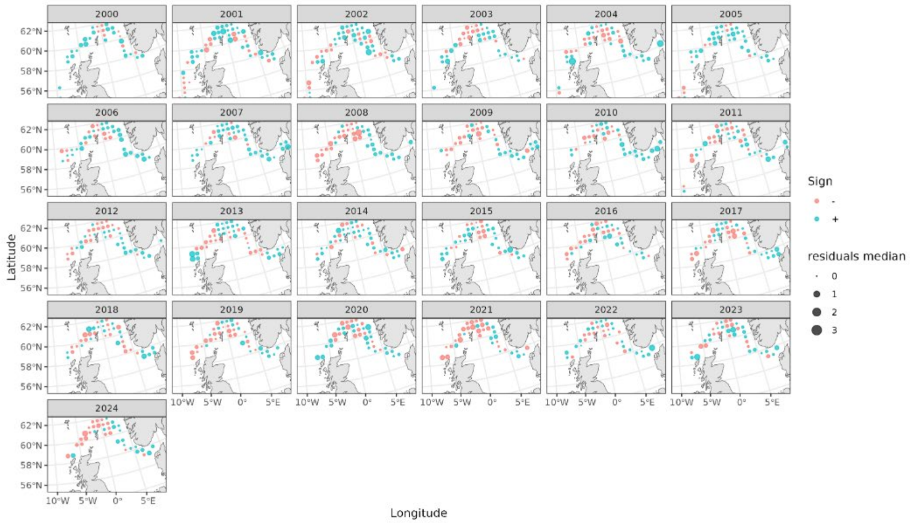

Figure 14.3.11. Saithe in subareas 4 and 6 and in Division 3.a. Maps of mean residuals from the CPUE index model per 0.5°x0.5° grid cell, per year (2000–2024).

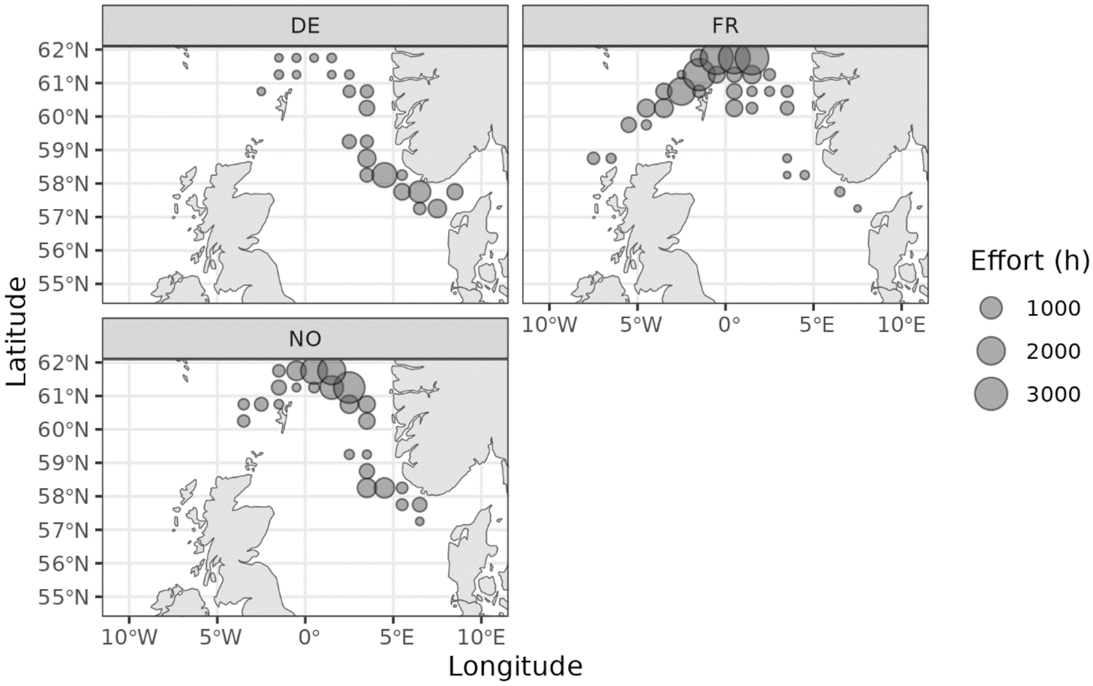

Figure 14.3.12. Spatial effort distribution, in 2024, of the three fleets contributing to the CPUE index.

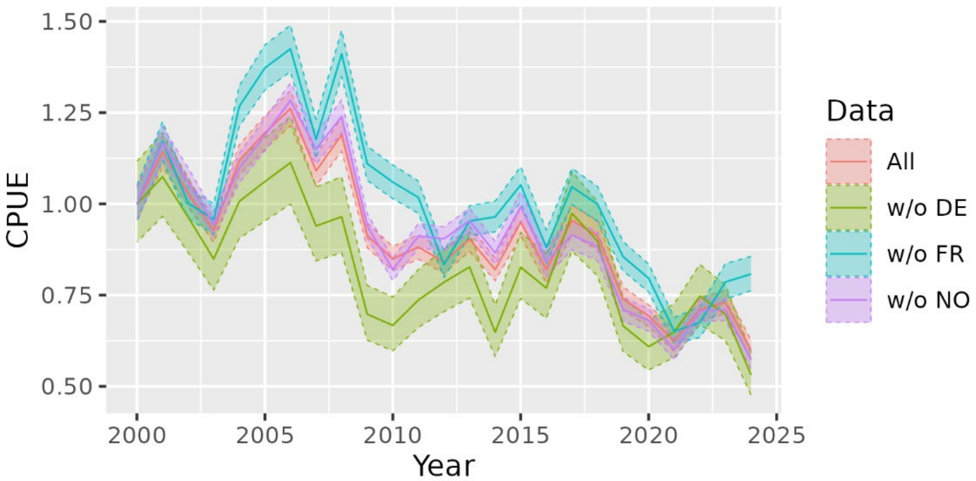

Figure 14.3.13. Leave one-nation-out runs with the GLM-based CPUE index model, as used in the assessment before 2024. CPUE is standardized to the first year of the time series. w/o = "without". DE = Germany, FR = France, NO = Norway.

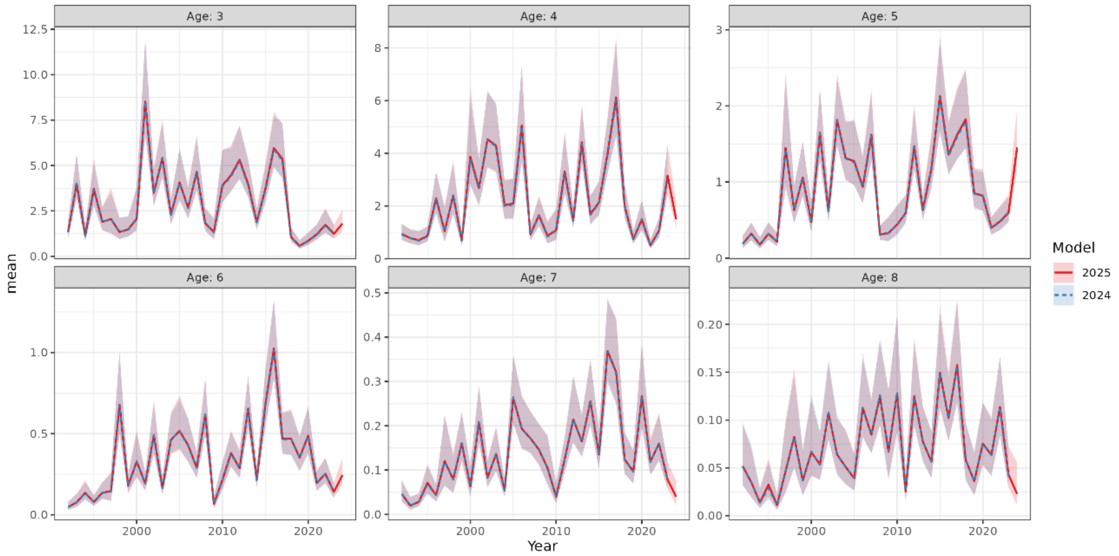

Figure 14.3.14. Saithe in subareas 4 and 6 and in Division 3.a. Relative standardized Q3–4 delta-GAM index of abundance-at-age, based on surveys in west of Scotland (Division 6.a), North Sea (Subarea 4) and Skagerrak (Division 3.a), for saithe ages 3–8, 1992–2024. Comparison of the updated index (red solid line, with 2024 data) to the previous 1992–2023 index, as used in the benchmark (blue dashed line).

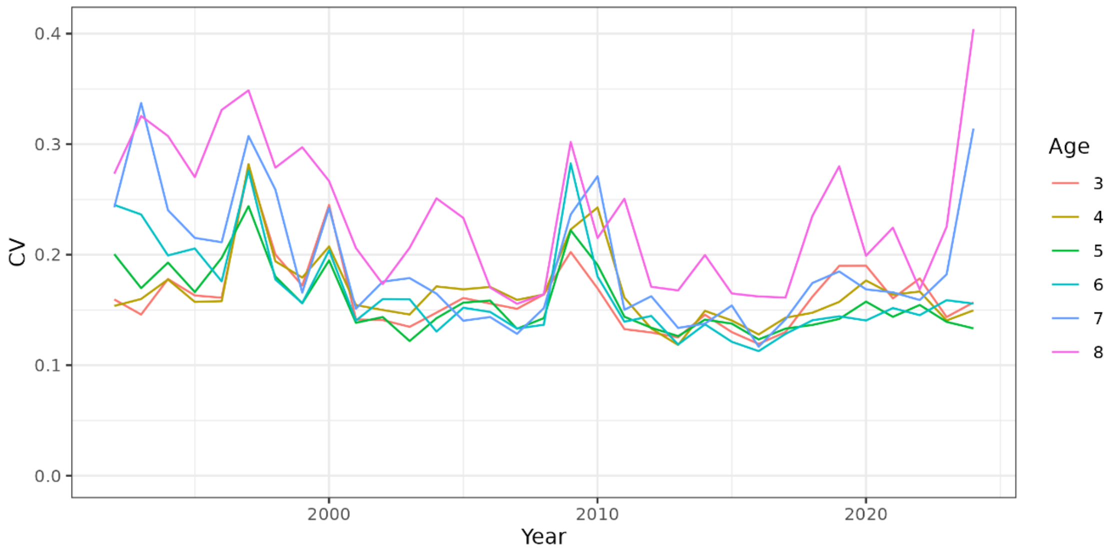

Figure 14.3.15. Saithe in subareas 4 and 6 and in Division 3.a. Yearly coefficients of variation of the standardized Q3–4 delta-GAM index of abundance-at-age, for saithe ages 3–8, 1992–2024.

Figure 14.3.16. Saithe in subareas 4 and 6 and in Division 3.a. Stacked standardized Q3–4 delta-GAM index of abundance-at-age 1–14+ per subareas, 1992–2024. Values are sum of predictions over grid cells in a Subarea, as used for the weighting of biological estimates at age per Subarea in the weighted mean per age group.

Figure 14.3.17. Saithe in subareas 4 and 6 and in Division 3.a. Relative abundance per Subarea at age 1–14+, 1992–2024. Based on the standardized Q3–4 delta-GAM index of abundance-at-age per subareas, age 1–14+, 1992–2024 (Figure 14.3.16).

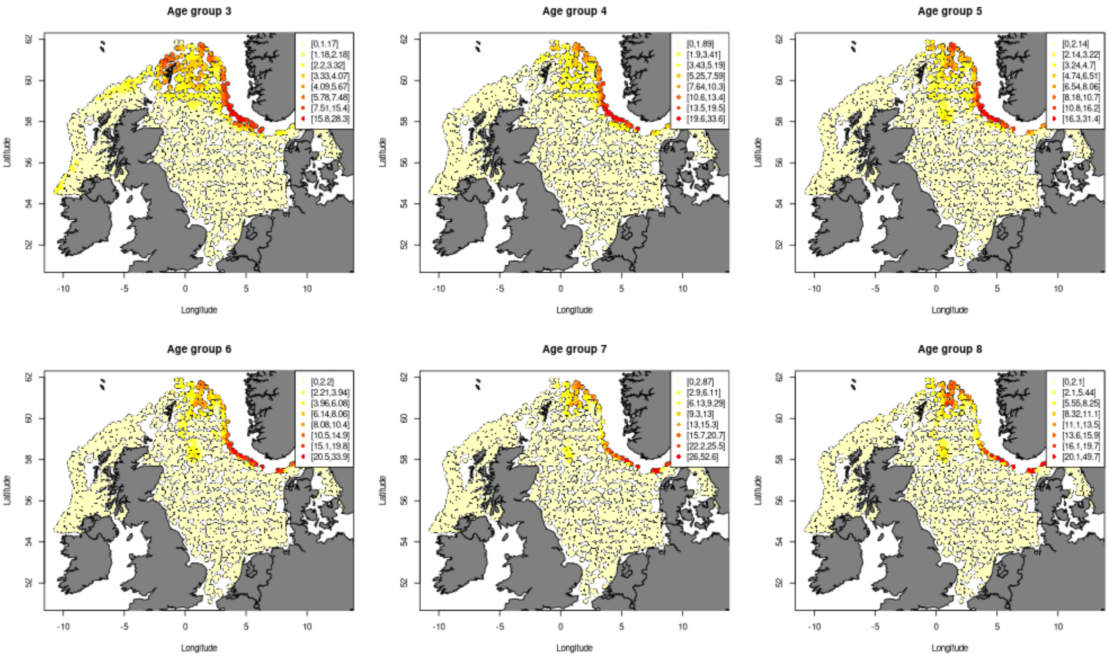

Figure 14.3.18. Spatial distribution in 2024 of Q3–4 delta-GAM index predictions (CPUEs) by age. Scales are different on each panel.

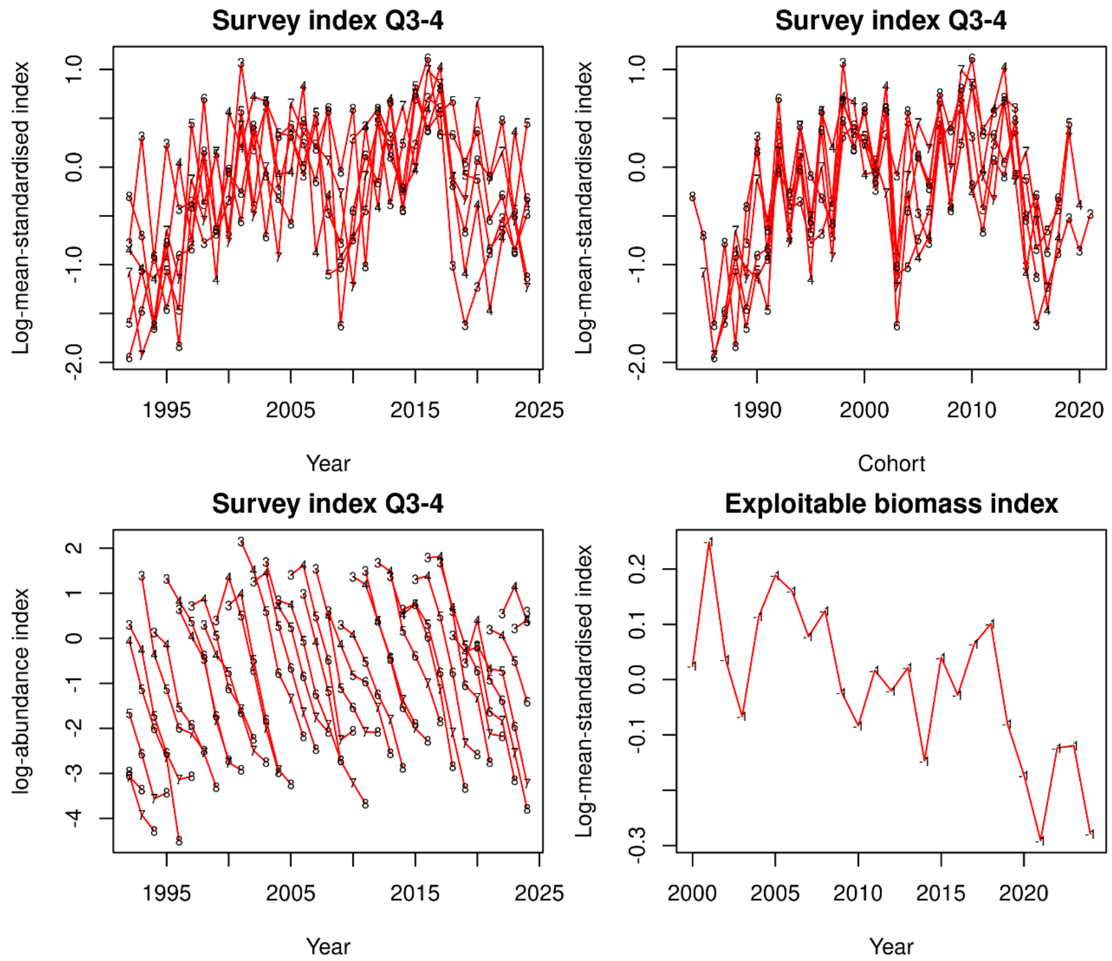

Figure 14.4.1. Saithe in subareas 4 and 6 and in Division 3.a. Research survey index, Q3–4, for ages 3 to 8, 1992–2024 is shown in terms of indices by age and year (top-left panel), indices by age and cohort (top-right panel), and log-catch curves by cohort (bottom-left panel). Commercial catch-per-unit-effort (CPUE) is shown in the bottom-right panel.

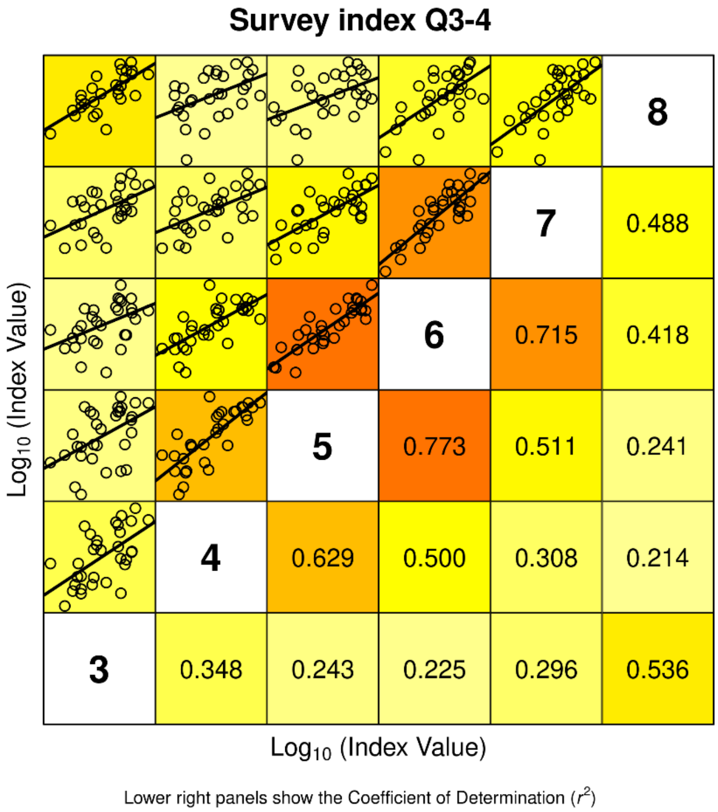

Figure 14.4.2. Saithe in subareas 4 and 6 and in Division 3.a. Internal consistencies for the survey index Q3–4, 1992–2024 ages 3 to 8.

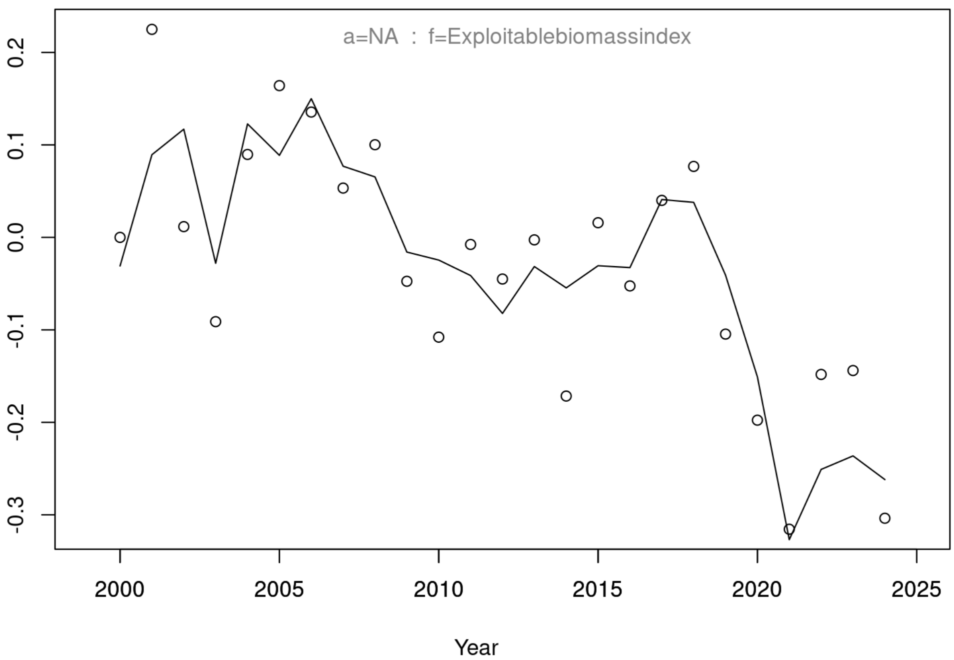

Figure 14.4.3. Saithe in subareas 4 and 6 and in Division 3.a. Standardized combined CPUE index (year effects, open circles) and fit of model after tuning to the exploitable biomass (solid line), 2000–2024.

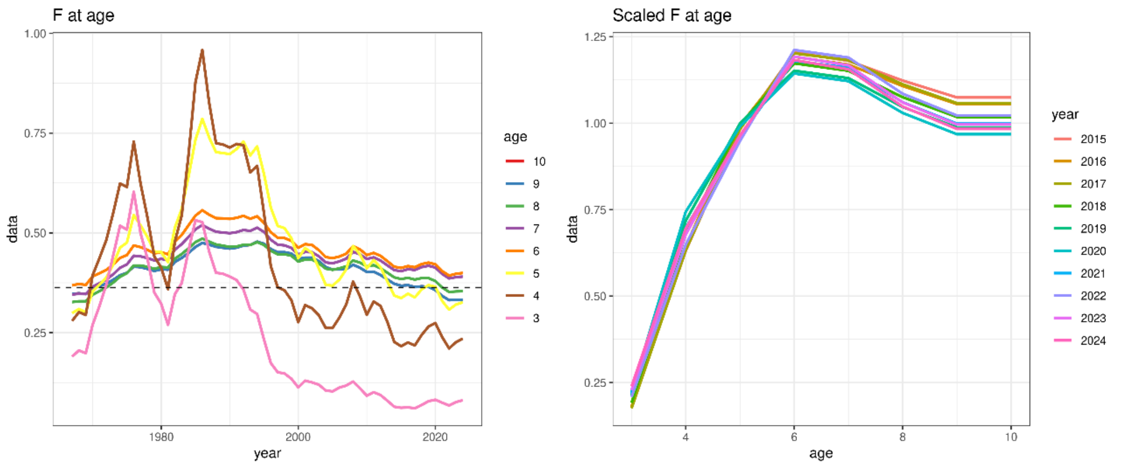

Figure 14.4.4. Saithe in subareas 4 and 6 and in Division 3.a. Fishing mortality-at-age for the final assessment model (1967–2024). Time-series (left panel) and scaled at $F_{4\text{–}7}$ for the last 10 years (right panel). F at age 9 and 10+, which are coupled in the model (9+), perfectly overlap.

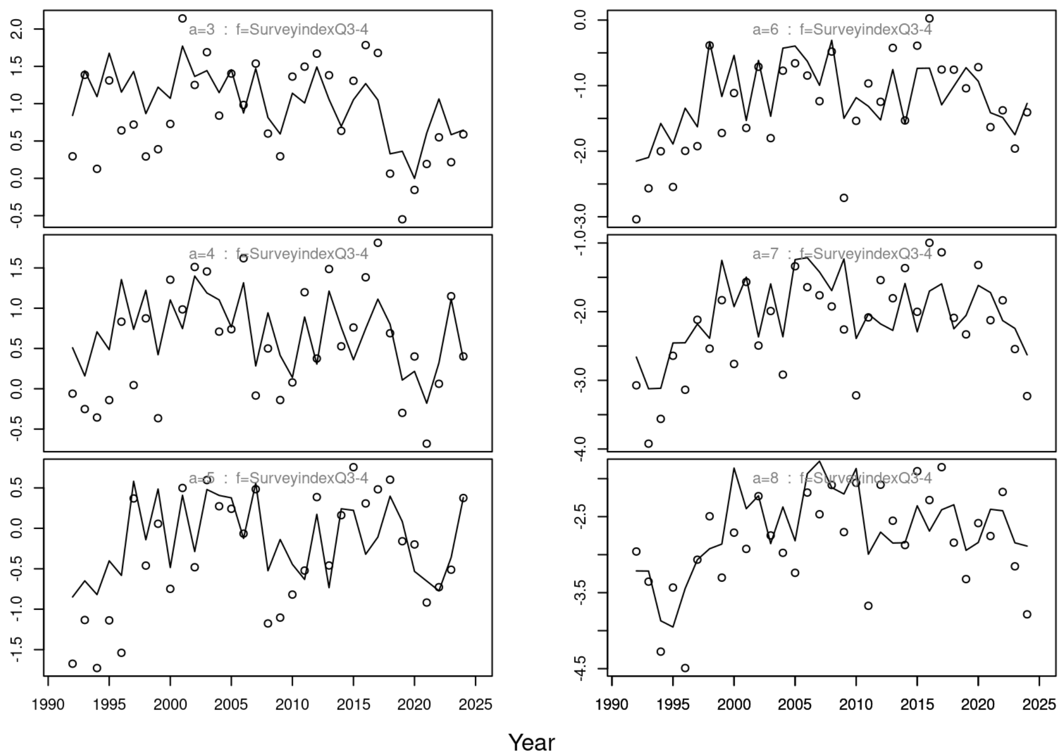

Figure 14.4.5. Saithe in subareas 4 and 6 and in Division 3.a. Survey Q3–4 index at age (age 3–8, open circles) and model fit (solid line), 1992–2024.

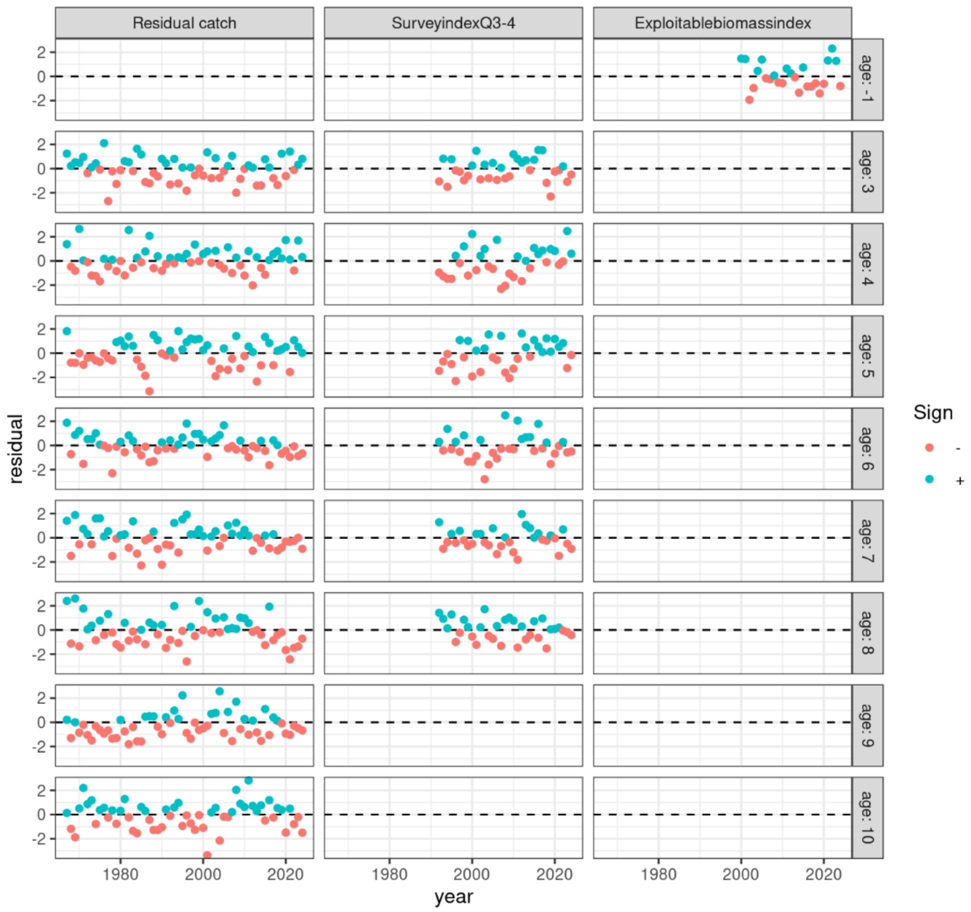

Figure 14.4.6. Saithe in subareas 4 and 6 and in Division 3.a. One-step ahead (serially independent) residual patterns of observations for the final SAM model. Plus-group: age 10+ for catch. Age: -1 indicates no age information.

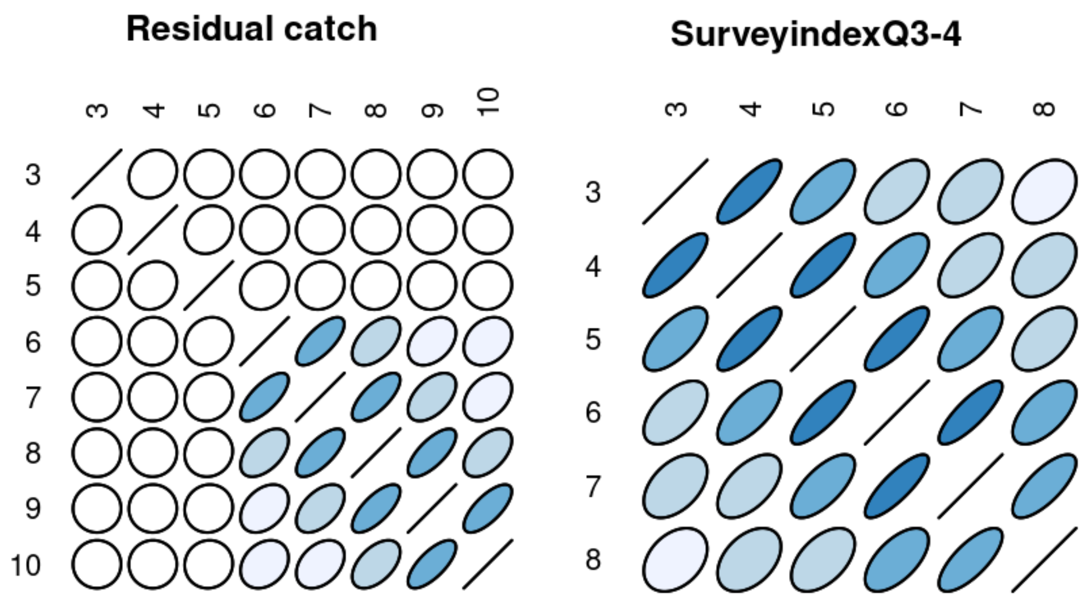

Figure 14.4.7. Saithe in subareas 4 and 6 and in Division 3.a. Correlation between age classes within years for catches (ages 3–10+; left panel) and survey index Q3–4 (ages 3–8; right panel) observations. The darker the blue colour, the stronger the positive correlation.

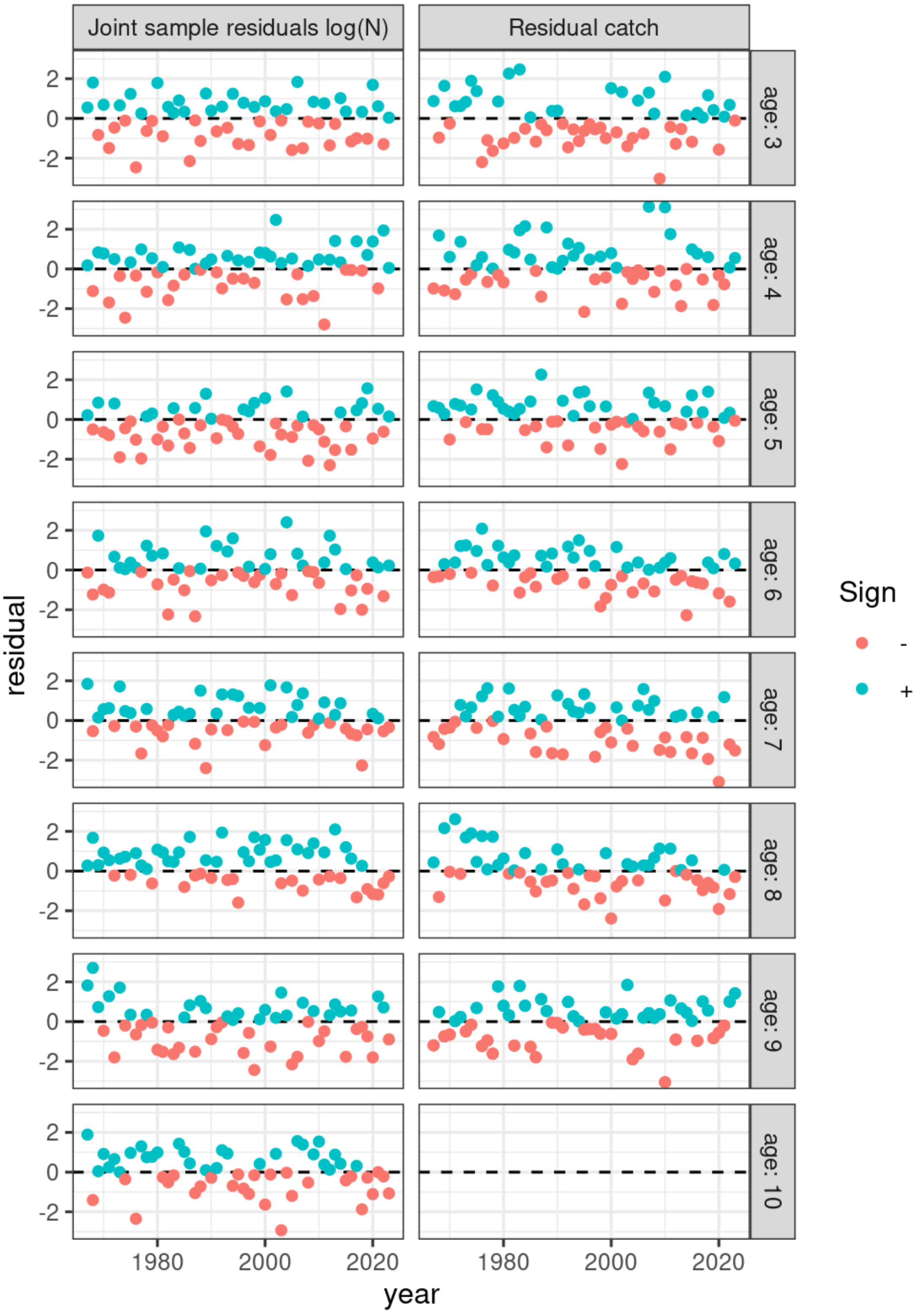

Figure 14.4.8. Saithe in subareas 4 and 6 and in Division 3.a. One-step ahead (serially independent) process error. Plus-groups: age 10+ for N, 9+ for catches (F process).

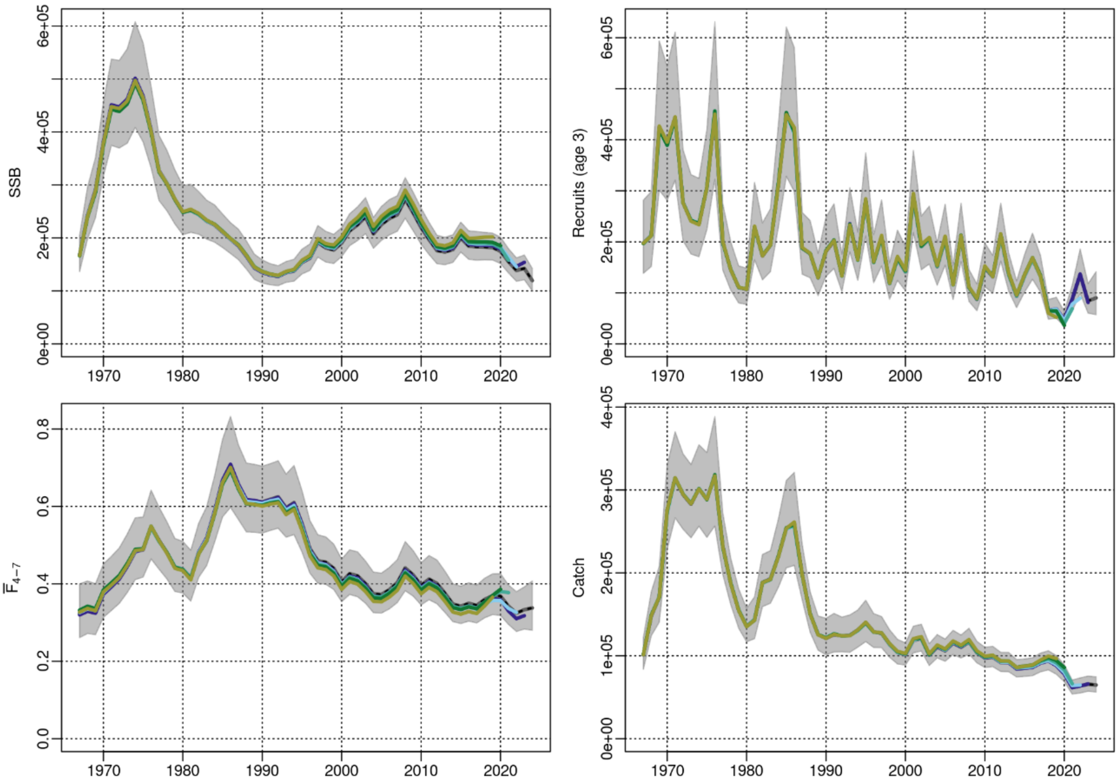

Figure 14.4.9. Saithe in subareas 4 and 6 and in Division 3.a. Five-year retrospective pattern in SSB, F_{4-7}, recruitment, and catches for the final assessment.

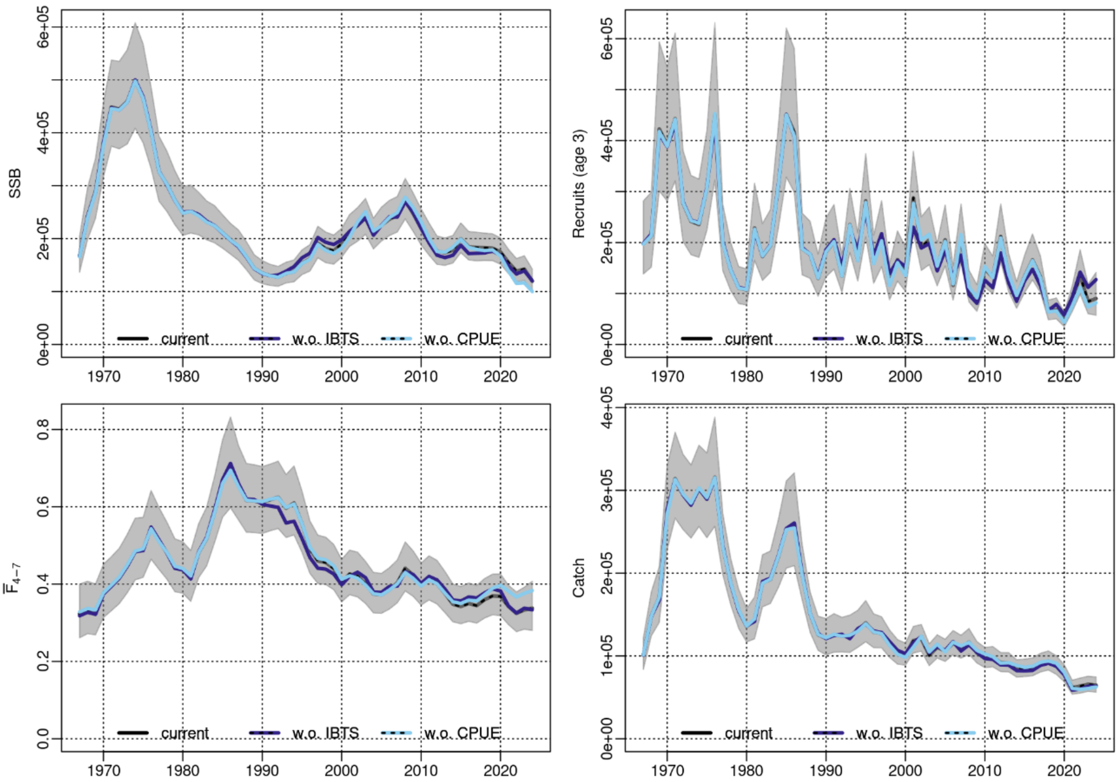

Figure 14.4.10. Saithe in subareas 4 and 6 and in Division 3.a. Stock summary of trends in SSB, F_{4-7}, recruitment, and catches for the final assessment model. Black lines and grey-shaded confidence interval indicate the final assessment model, including the survey Q3–4 indices for ages 3–8 and the CPUE index. The light blue line is the assessment with only the survey Q3–4 tuning series, while the dark blue one is the assessment with only the CPUE index.

Figure 14.5.1. Saithe in subareas 4 and 6 and in Division 3.a. Summary of stock assessment in relation to reference points for SSB and F. Predicted recruitment values are light shaded. Shaded areas (F, SSB) and error bars (R) indicate pointwise 95% confidence intervals. SSB estimate in 2024 (diamond) relies on the stochastic recruitment STF for age 3 (8% of SSB); ages 4–10+ are 2024 survivors.

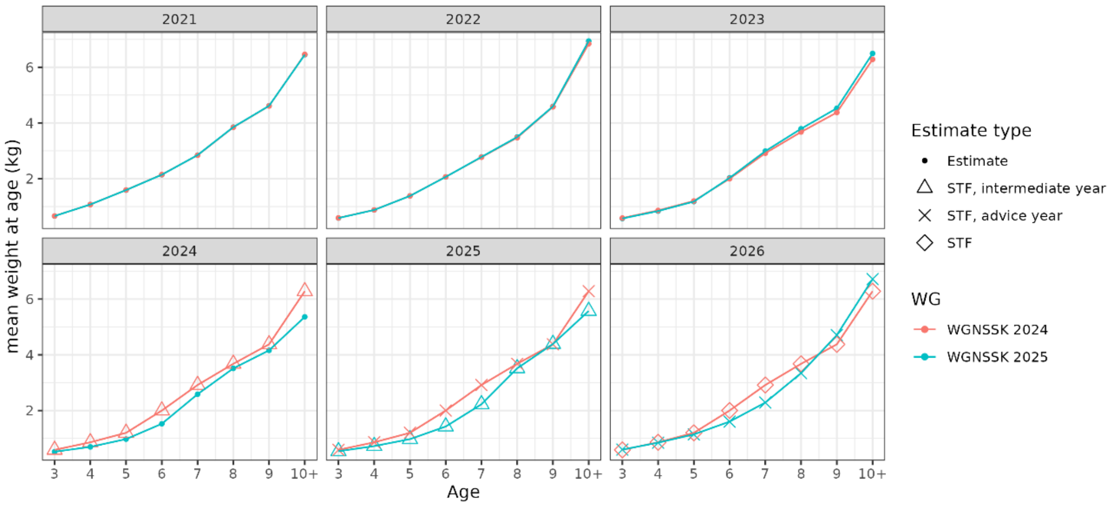

Figure 14.7.1. Saithe in subareas 4 and 6 and in Division 3.a. Comparison of the 2021–2026 mean stock weights-at-age (InterCatch estimates and forecast assumptions) between the current assessment (WGNSSK 2025) and the previous assessment (WGNSSK 2024).

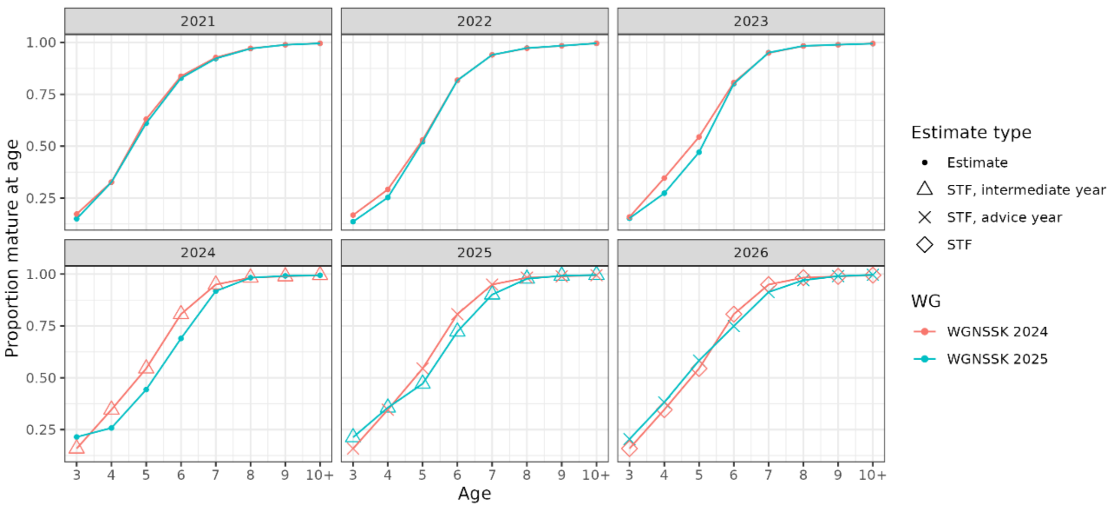

Figure 14.7.2. Saithe in subareas 4 and 6 and in Division 3.a. Comparison of the 2021–2026 proportion mature-at-age (estimates and forecast assumptions) between the current assessment (WGNSSK 2025) and the previous assessment (WGNSSK 2024), both model-based.

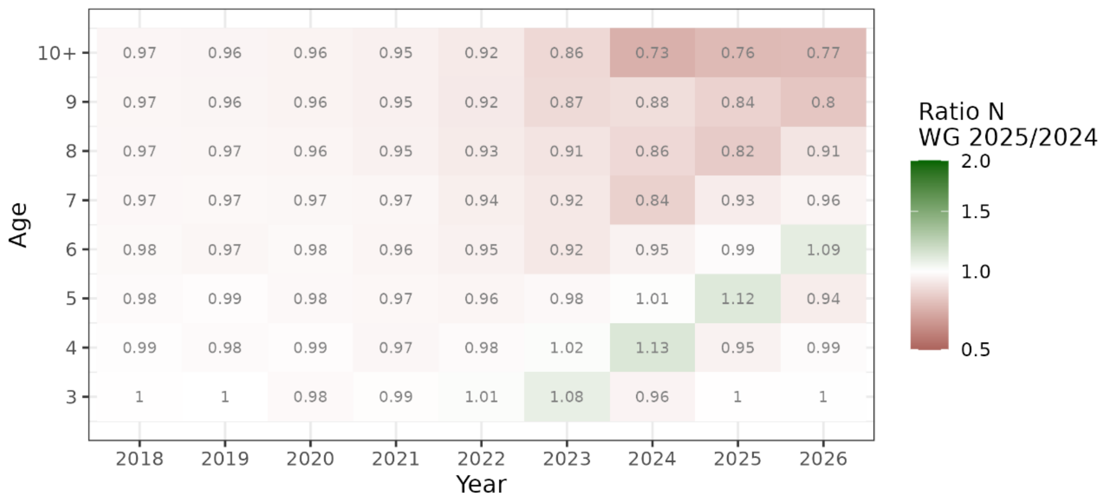

Figure 14.7.3. Saithe in subareas 4 and 6 and in Division 3.a. Ratio of number-at-age estimates (model and forecast) between the current assessment (WGNSSK 2025; intermediate year 2025, STF 2026) and the previous assessment (WGNSSK 2024; intermediate year 2024, STF 2025–2026) for the period 2018–2026. A value over 1 indicates an increased estimate.

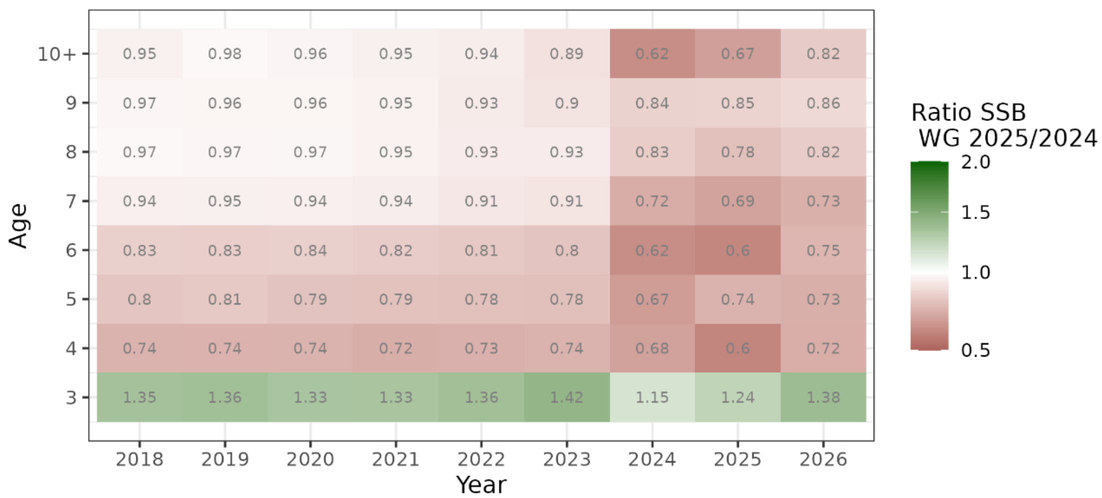

Figure 14.7.4. Saithe in subareas 4 and 6 and in Division 3.a. Ratio of spawning-stock biomass (SSB)-at-age estimates (model and forecast) between the current assessment (WGNSSK 2025; intermediate year 2025, STF 2026) and the previous assessment (WGNSSK 2024; intermediate year 2024, STF 2025–2026) for the period 2018–2026. A value over 1 indicates an increased estimate.

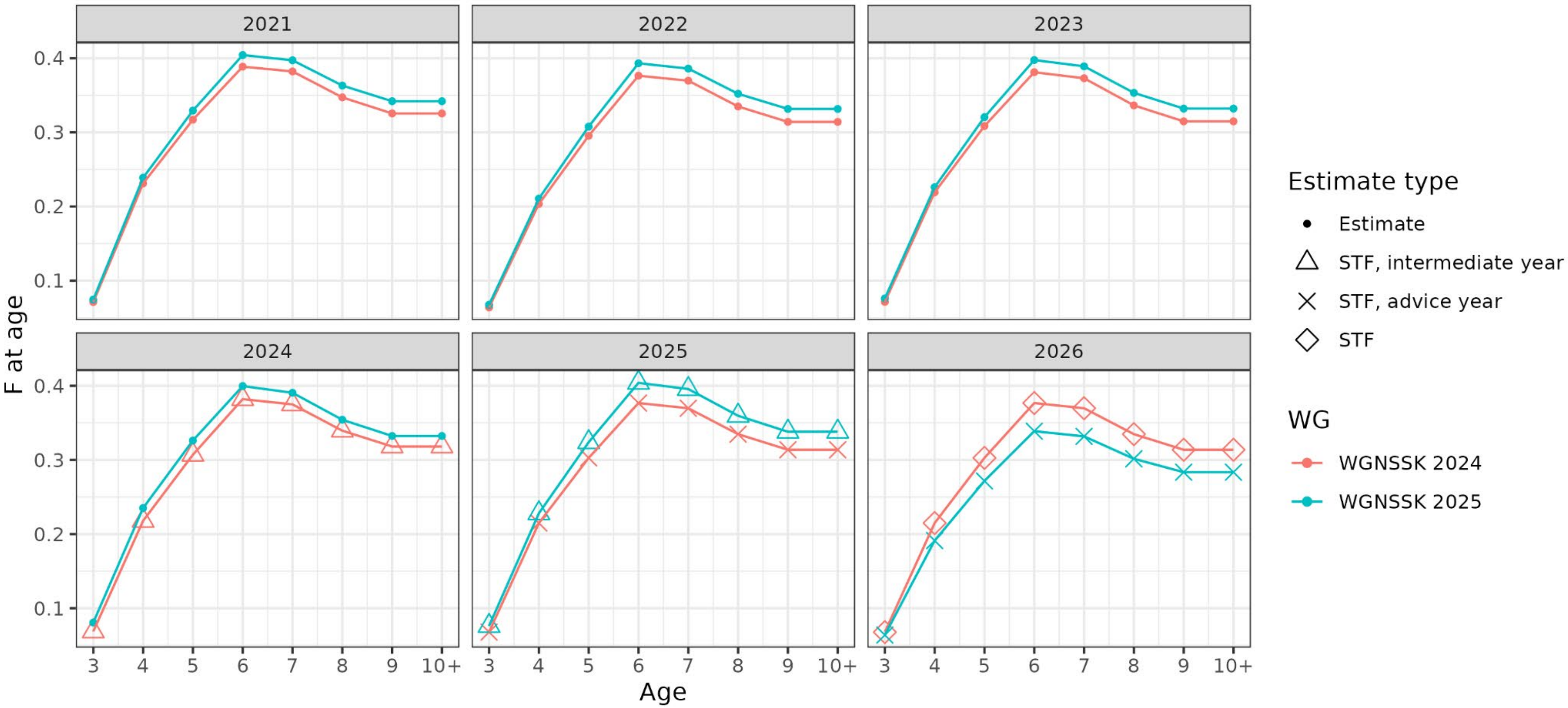

Figure 14.7.5. Saithe in subareas 4 and 6 and in Division 3.a. Comparison of fishing mortality (F)-at-age estimates per year (model and forecast) between the current assessment (WGNSSK 2025) and the previous assessment (WGNSSK 2024).

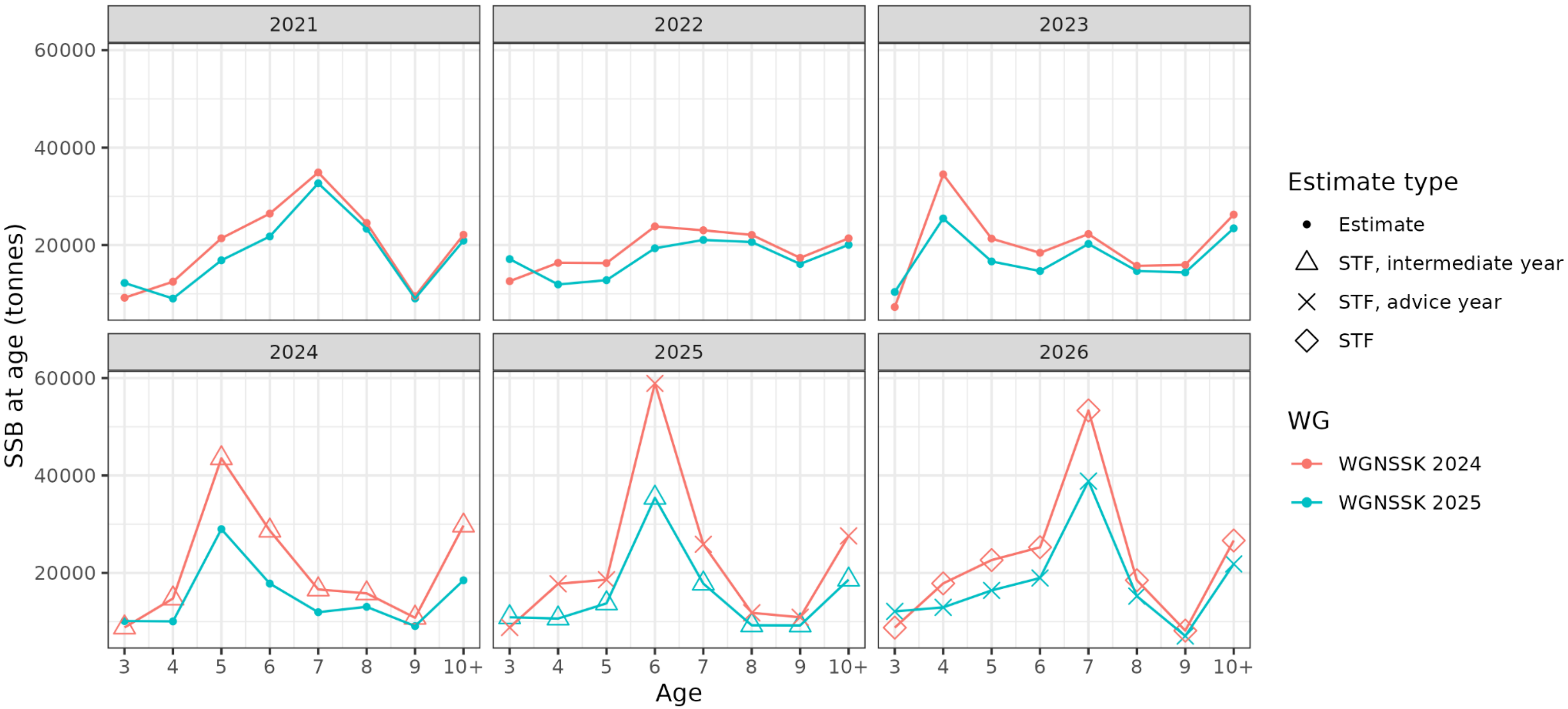

Figure 14.7.6. Saithe in subareas 4 and 6 and in Division 3.a. Comparison of spawning-stock biomass (SSB)-at-age estimates per year (model and forecast) between the current assessment (WGNSSK 2025) and the previous assessment (WGNSSK 2024).

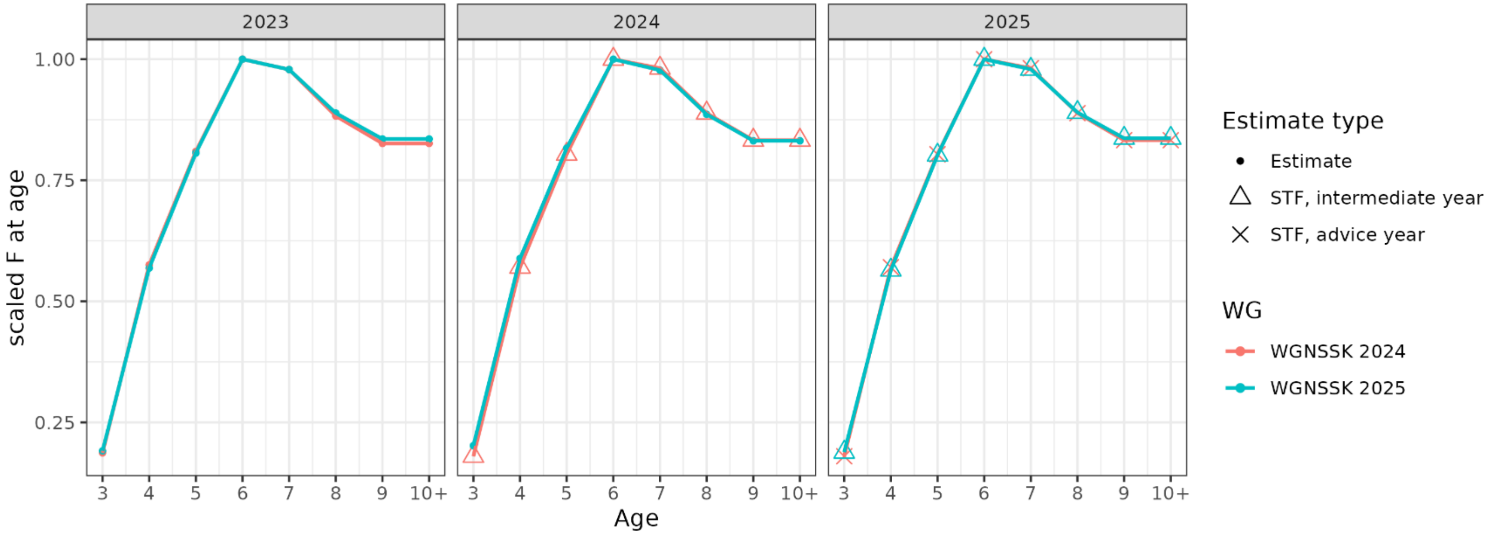

Figure 14.7.7. Saithe in subareas 4 and 6 and in Division 3.a. Comparison of the last three years of scaled fishing mortality (F) ogives (model and forecast) between the current assessment (WGNSSK 2025) and the previous assessment (WGNSSK 2024).

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